



**Verified Carbon
Standard**

ÖDEMiŞ INTEGRATED SOLID WASTE MANAGEMENT PLANT PROJECT



Document Prepared by Life İklim ve Enerji Ltd. Şti.

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1 PROJECT DETAILS

1.1 Summary Description of the Project

İzmir Metropolitan Municipality has implemented the İzmir Integrated Waste Management Plan to manage municipal solid waste generated in all of its districts in 2018. The Metropolitan Municipality determined five sub-management regions within the Integrated Waste Management Plan's scope. The "Güney (south)-1 Sub-Management Region" where this Project serves consists of 7 districts in total, namely Ödemiş, Kiraz, Beydağ, Bayındır, Tire, Selcuk and Torbalı. In line with this Management Plan, İzmir Metropolitan Municipality constructed an Integrated Solid Waste Treatment plant in Ödemiş District to treat the generated waste in these seven districts.

İzmir Metropolitan Municipality has appointed Biosun Kati Atik İşleme Enerji ve Çevre San. Tic. A.Ş. (hereafter referred to as the Project Proponent) to construct and operate Ödemiş Integrated Solid Waste Management Plant for 29 years in Ödemiş District, İzmir Province, Turkey. Different waste treatment processes are applied to the incoming fresh waste in the Plant. Ödemiş Integrated Solid Waste Management Plant (hereafter referred to as Ödemiş ISWMP) includes Mechanical/Thermal Treatment to produce Refuse-Derived Fuel (RDF) and Anaerobic Digestion with Biogas Recovery. Ödemiş ISWMP aims to regain packaging and biodegradable wastes for the economy, generate energy, and direct residual wastes to landfills. In the Plant, packaging wastes and valuable metals are recycled, methane gas is obtained from biodegradable wastes by anaerobic digestion and then converted into electricity and heat in the cogeneration plant. Some portion of the digestate from the anaerobic digesters is treated further to produce organic fertiliser, and the remaining part is directed to the thermal drying unit, where the moisture content of the digestate is reduced. The portion of digestate with reduced moisture content is then directed to the landfill.

The Ödemiş ISWMP accepts a daily mixed fresh waste of 845 tons/day. The incoming mixed waste is first treated in the Material Recovery Facility, where recyclable, biodegradable, bulky, combustible, and residual wastes are separated. Biodegradable wastes resulting from Material Recovery Facility are mixed with the separately collected biodegradables, agricultural&forestry wastes from gardens and parks, and animal farms (i.e. animal manure). The mixed biodegradable wastes are then forwarded to the anaerobic digesters, where a total daily waste amount of 800 tons is processed. The biogas that is released during the biodegradation of organic wastes in anaerobic digesters is used for electricity and heat production in the cogeneration unit.

On 03/10/2020, EMRA granted the Project activity with a Generation License of 20.28 MWe for 27 years. The License is granted for the Plant to have 8 digesters and 13 gas engines for a capacity of 20.28 MWe. The plant's design also considered an incoming waste amount

for the year 2050. There are currently 3 digesters and five engines operating in the plant, with a total installed capacity of 7.8 MWe. The commissioning date of the first two engines is 29/12/2020 with 1,560 kWe each, total electrical power of 3,120 kWe. The other three engines, each with an electrical power of 1,560 kWe, were commissioned on 14/10/2021 with a total electrical power of 4,680 kWe. Following the Capacity Report of the Project Activity for an installed capacity of 7.8 MWe, the biogas plant in the Ödemis ISWMP can produce 13,660,714 m³ of biogas yearly and accordingly; the Plant is expected to produce 32,307,586 kWh of gross electricity annually. The construction of the power plant was started in February 2020 with the Construction Agreement signed with the contractor company, and the Plant started generating electricity on 29/12/2020.

Before the project activity, the municipal solid waste, agricultural and forestry waste were collected and disposed of in the municipal landfill sites, which do not have any measurements to collect and destroy the landfill gas. The generated wastes from the seven districts were disposed of in two landfill sites. The two municipal landfill sites were located in Harmandalı and Bergama Districts in İzmir. The two municipal landfill sites have been operational since 1996. Before the project activity, there wasn't any recycling activity for municipal mixed waste. The proposed Project doesn't account for the emission reduction due to recycling plastic, glass or metals. Moreover, without the project activity, the animal manure would be treated in the uncovered anaerobic lagoons, the electricity generated by the Project would be replaced by the power plants connected to the national grid, and fossil fuel fuel-fired boilers would supply the thermal energy.

1.2 Sectoral Scope and Project Type

Sectoral scope: 1 Renewable Energy

Sectoral scope: 13 Waste Recovery and disposal

Type I: Renewable Energy Projects

Type III: Other project activities

The Project is neither an AFOLU nor a grouped project

1.3 Project Eligibility

As per the applied methodology ACM0022 ver. 3.0 - Alternative waste treatment processes *"This methodology covers project activities implementing and operating new plants for the treatment of fresh waste that would otherwise be disposed in a solid waste disposal site (SWDS)."*

The project activity involves implementing and operating waste treatment plants, without which wastes generated would be disposed of in a solid waste disposal site (SWDS). Therefore the project activity is eligible.

As per the applied methodology ACM0010 ver. 8.0 - GHG emission reductions from manure management systems

"This methodology applies to project activities that include destruction of methane emissions and displacement of a more GHG-intensive service in manure management of livestock farms by introducing a new animal waste management system or a combination of animal waste management systems that result in less GHG emissions."

The project activity destroys methane emissions in livestock farms by treating the collected manure in anaerobic digesters. Therefore the project activity is eligible.

1.4 Project Design

- ☒ The Project includes a single location or installation only
- ☐ The Project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☐ The Project is a grouped project

Eligibility Criteria

The Project Activity is not a grouped project. The Project Activity belongs to one project proponent. In the scope of this project activity, only one location will be considered for emission reductions.

1.5 Project Proponent

Organisation name	Biosun Ödemiş Kati Atık İşleme Enerji ve Çevre San. Tic. A.Ş.
Contact person	Mr Gültekin Özkan
Title	Plant Manager
Address	Emmioglu Mah. 2. Şeytanlı Karaagac Mevkii No:16/1 Ödemiş/İZMİR
Telephone	+90 (232) 343 58 35
Email	gultekinozkan@biosunas.com.tr

1.6 Other Entities Involved in the Project

Organisation name	Life İklim ve Enerji Ltd. Şti.
Role in the Project	Consultant for Carbon Certification
Contact person	Mr. Ramazan Aslan
Title	General Manager
Address	Oğuzlar Mah. 1377. Sk. No:19/9 06520, Balgat, Çankaya/Ankara
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Email	ramazan.aslan@lifeenerji.com

1.7 Ownership

Ownership of the Ödemiş ISWMP project belongs to Biosun Ödemiş Kati Atik İşleme Enerji ve Çevre San. Tic. A.Ş., and all emission reductions/removals are granted to the company. The tax plate as proof of operating certification of the company given by the relevant tax office of the host country is provided to the validating VVB as proof of the Title.

1.8 Project Start Date

The project start date is determined by the earliest commissioning date of the biogas engines. The start date of the project activity is 29-December-2020, i.e. the provisional acceptance protocol of the first engine.

1.9 Project Crediting Period

The Project crediting period for this Project is 7 years, renewable twice.

First Crediting Period Start Date: 29/12/2020

First Crediting Period End Date: 28/12/2027

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the Project are:

☐ <20,000 tCO₂e/year

☐ 20,000 – 100,000 tCO₂e/year

☒ 100,001 – 1,000,000 tCO₂e/year

☐ >1,000,000 tCO₂e/year

Project Scale	
Project	
Large Project	X

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
2021	63,804
2022	94,486
2023	121,254
2024	144,733
2025	165,393
2026	183,691
2027	199,948
Total estimated ERs	973,310
Total number of crediting years	7
Average annual ERs	139,044

1.11 Description of the Project Activity

As per Izmir Integrated Waste Management Plan, all seven districts' rural and urban areas are provided with collection containers for mixed waste and recyclables. Waste collection vehicles are used to collect mixed waste and recyclables from containers. The collected waste is brought to the transfer stations located in Çaylı, Torbalı, Kısıkköy and Selcuk Transfer Stations, and from here, waste is transferred to the Ödemiş ISWMP to be treated.

The Ödemiş ISWMP project's design includes:

- Municipal Solid Waste reception unit
- Material Recovery Facility

- Biogas Plant
 - Biodegradable Wastes Reception Units
 - Feed optimisation and pumping system
 - Anaerobic Digesters
 - Desulphurisation Unit
 - Cogeneration Unit and Flare Unit
 - Hygienization and Dewatering Unit
 - Storage Tanks
- Refuse-derived Fuel Plant

Material Recovery Facility

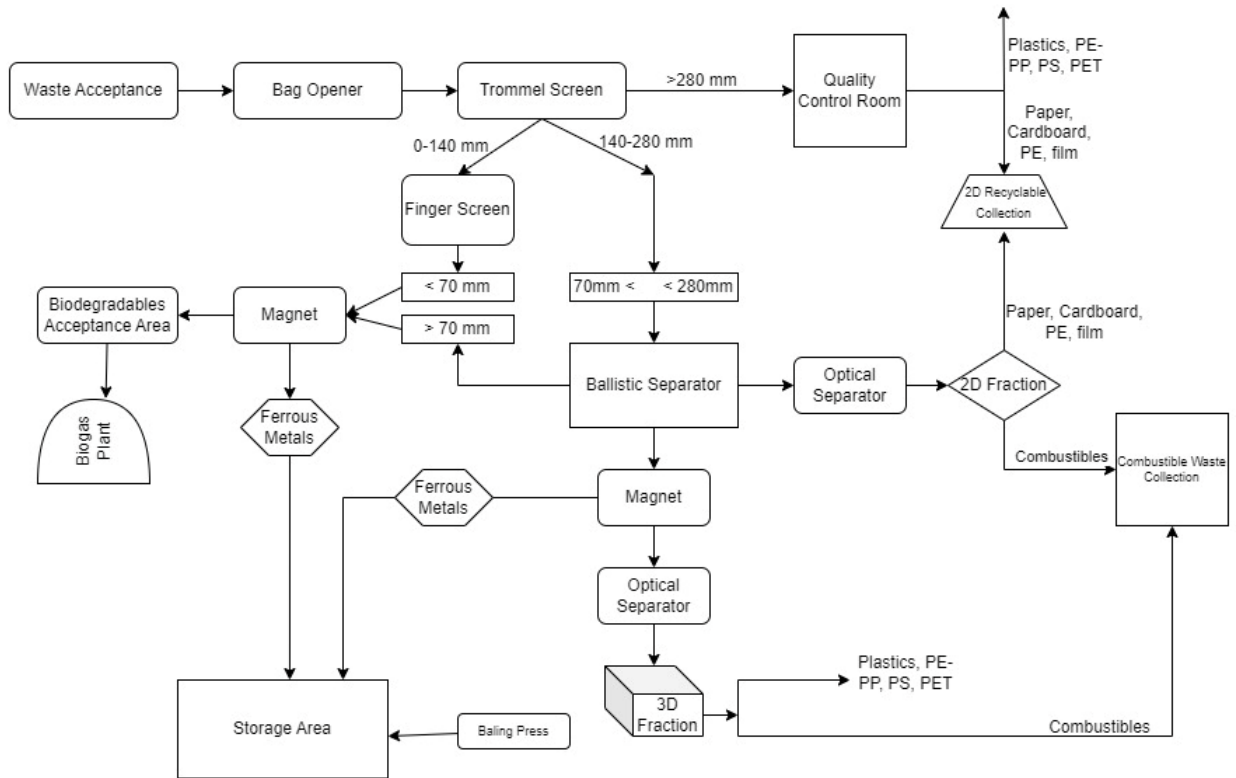
The Material Recovery Facility consists of easily operational and maintainable sorting machinery with a design capacity of 718 tons per day. The basic process flow of the Material Recovery Facility consists of waste reception leading to a feeding bunker & bag opener by means of a wheeled loader. After the bag opener, the fresh waste is forwarded to a trommel screen. Trommel screens are the machines which separate the waste according to predetermined sizes. This process occurs with the help of the drum inside of the trommel screen, which has holes with a specific diameter. The three different sizes of waste are obtained from the trommel screen: the waste with a particle size greater than 280 mm, the waste with a particle size between 140-280 mm and the waste with a particle size of 0-140 mm. The recyclable part of waste with a particle size greater than 280 mm is recovered, and the non-recyclables are sent to the landfill.

The rejects with a diameter of 0-140 mm from the trommel screen are then processed towards the finger screens. Finger screens are named after finger-shaped steel pipes. Springs in the finger screen separate the material via vibration into two different sizes. Rejects with a smaller particle size from the finger screen are then processed towards the magnetic separator. Magnets are used to separate ferrous materials from the rest of the refuse. A belt magnet is installed above a conveyor belt, with the belt on the magnet moving across the conveyor carrying the refuse. And after magnets the wastes are processed towards the Biogas Plant.

The rejects with a diameter of 70-280 mm from the trommel screen are then processed towards a ballistic separator. A ballistic separator is a mechanical sorting device with oscillating paddles. Alternating sets of paddles provide agitation of the material stream. The agitation has a different effect on the mixed waste stream, causing a separation of rigid items off one end and flexible items off the other. After the ballistic separator, magnets and optical sorters are employed to recover glass, plastics and ferrous metals. Moreover, baling presses are used in the Plant to bale materials such as Paper, paper, Nylon, Plastic, wood, sack, tin, fibre, and textile waste. All non-recyclable combustible wastes are then forwarded to the

Refuse-derived fuel plant to produce fuel, and all bulky wastes collected in the Plant are sent to the landfill.

The Material Recovery Facility flow scheme is shown in the figure below.



Biogas Plant (Anaerobic Digesters)

Three anaerobic digesters operate in the biogas plant with 400 tons/day capacity. The reactors are made from reinforced concrete, leak-proof, and designed to have a self-heating and mixing system. Three different waste reception units are employed in the Plant for biodegradable wastes from the mechanical treatment, animal manure collected from the animal farms and the separately collected garden park wastes. The different types of biodegradable wastes are then homogenised in the unit to be fed to the anaerobic digester. There are two separate homogenisation tanks in the biogas plant. Biodegradable wastes from the Material Recovery Facility and separately collected biodegradable wastes, animal manure, agricultural&forestry wastes are accepted to all three anaerobic digesters after homogenisation. Biodegradable waste is then digested by microorganisms under anaerobic conditions and converted into carbon dioxide and methane gas. On average, the hydraulic retention time of the waste in the digesters is 20 days. In the digester, methane gas is produced under mesophilic conditions at 38 degrees Celsius; the methane gas produced is withdrawn by blowers from the digester and delivered to the cogeneration unit to obtain electrical and thermal energy. Hot water pipes placed inside the digesters provide automatic

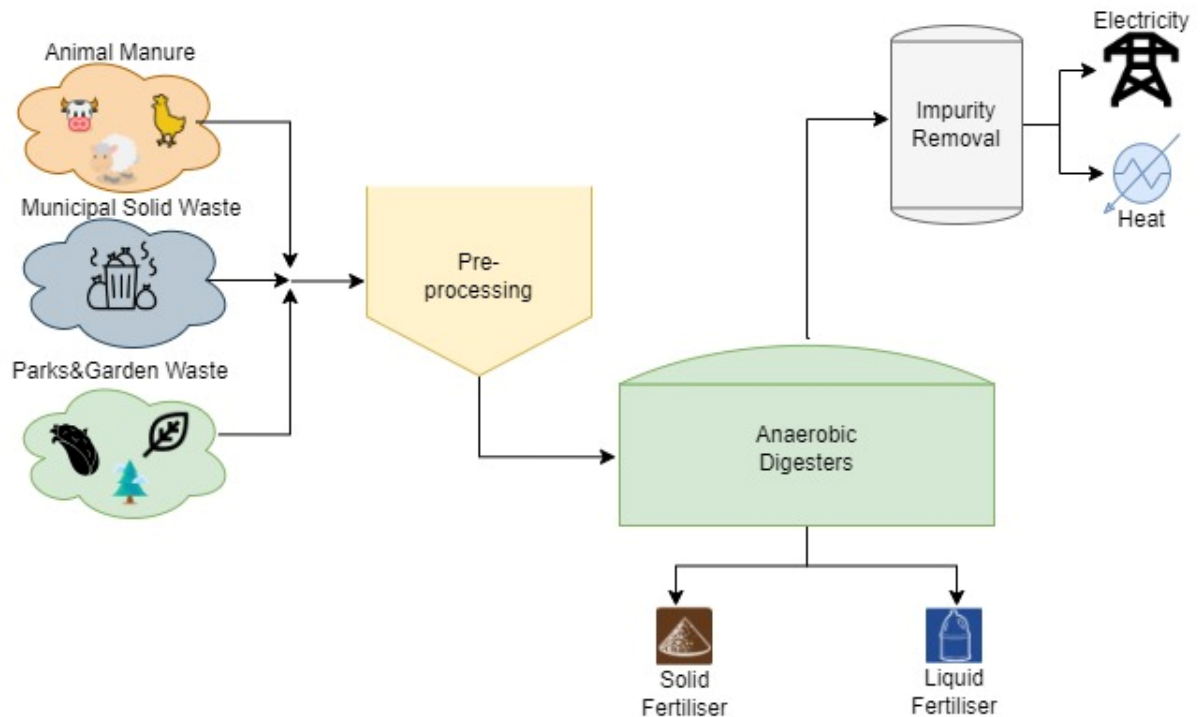
control of the temperature of the digester to keep it constant with continuous hot water circulation.

Also, ten mixers inside the digesters are activated according to the time parameters determined in the Plant's automation system to keep the mixture homogenous.

Before starting the project activity (in the baseline scenario), manure from the farms was being washed out from underneath the animal barns and stored in anaerobic lagoons, which allowed the manure to decay and emit methane into the atmosphere along with several other toxic gases. Such applications also bear the risk of lagoon overflow due to increased rainfalls, strong Biogas or improper construction.

Energy requirements for the Plant will be supplied from the national grid, and a generator will be ready to use in an emergency electricity need.

The Project is estimated to generate a net 32,307.586 MWh/y of electricity, while the biogas production capacity will be around 13,660,714 m³/y¹.



Desulphurisation Unit

Hydrogen sulphide, a highly corrosive acidic gas contained in the biogas, and water vapour are also eliminated in the desulphurisation unit employed in the biogas plant. Hydrogen sulfide (H₂S) in the biogas is reduced to 150 ppm by desulphurisation. After the

¹ Please See Capacity Report of the Project Activity

desulphurisation unit, the produced biogas is stored in double membrane inflatable roofs and then accelerated by a blower system to the (CHP unit) cogeneration building.

Cogeneration Unit and Flare Unit

The CHP unit has five electricity-generating engines, each with a capacity of 1.56 MWe totalling up to 7.80 MWe. Biogas energy fed to gas engines will be converted into electricity with 42.1% efficiency and thermal energy with 42.1% efficiency. All the electricity produced is supplied to the grid, which also meets the electricity consumed by the Plant. The total efficiency of the gas engines used is over 84%. There is also a flare unit with a capacity of 6000m³/hr in the biogas plant for emergencies where the engines don't combust produced biogas. The combustion temperature of the flare is around 800-900°C, and the flare's efficiency is higher than 95%.

The heat generated in the CHP plant is used to heat the anaerobic digesters, the administrative building, and the thermal drying unit in the Refuse-derived fuel plant.

Hygienization and Dewatering Unit

The digestate from the digesters is sent to the dewatering unit. The digestate with a dry matter of 20% is increased to 40% at the dewatering unit. And the dry matter content of the remaining digestate is reduced to less than 10% producing a liquid fertiliser. The solid fertiliser from the dewatering unit is then subjected to the hygienization process at a temperature of 70 degrees for 1 hour per the regulation and then sent to the landfill. The liquid fertiliser from the dewatering unit is stored in a closed equalisation tank with a capacity of 500 tons and recycled back to the digesters. In the later stages, the excess liquid fertiliser will be subjected to pre-treatment in the Plant's wastewater treatment plant to be established and discharged to the İzmir Metropolitan Municipality's wastewater treatment plant.

Refuse-derived Fuel Plant

The non-recyclable, combustible wastes sorted in the Material Recovery Facility are utilised in the RDF Plant. The wastes utilised in the RDF Plant include textile wastes, fibre, surface coating materials, diapers, pads, sawdust, splinters, wood, plywood, etc. The RDF Plant's capacity is designed considering an incoming amount of 476 tons/day of waste.

Part of the solid fertiliser produced in the biogas plant is further dried and treated in the RDF plant. Thus its calorific value can be increased to be sent to the cement factories for incineration. The wastes to be used as RDF is first reduced in size by shredders and dried in the RDF plant. According to the waste characterisation study conducted by the Municipality, combustible wastes have a percentage of %25.30 of all the incoming waste to the ISWMP. The residual waste resulting from the RDF Plant is sent to the landfill.

The flow scheme of the Biogas Plant and RDF Plant can be seen in the below figure.

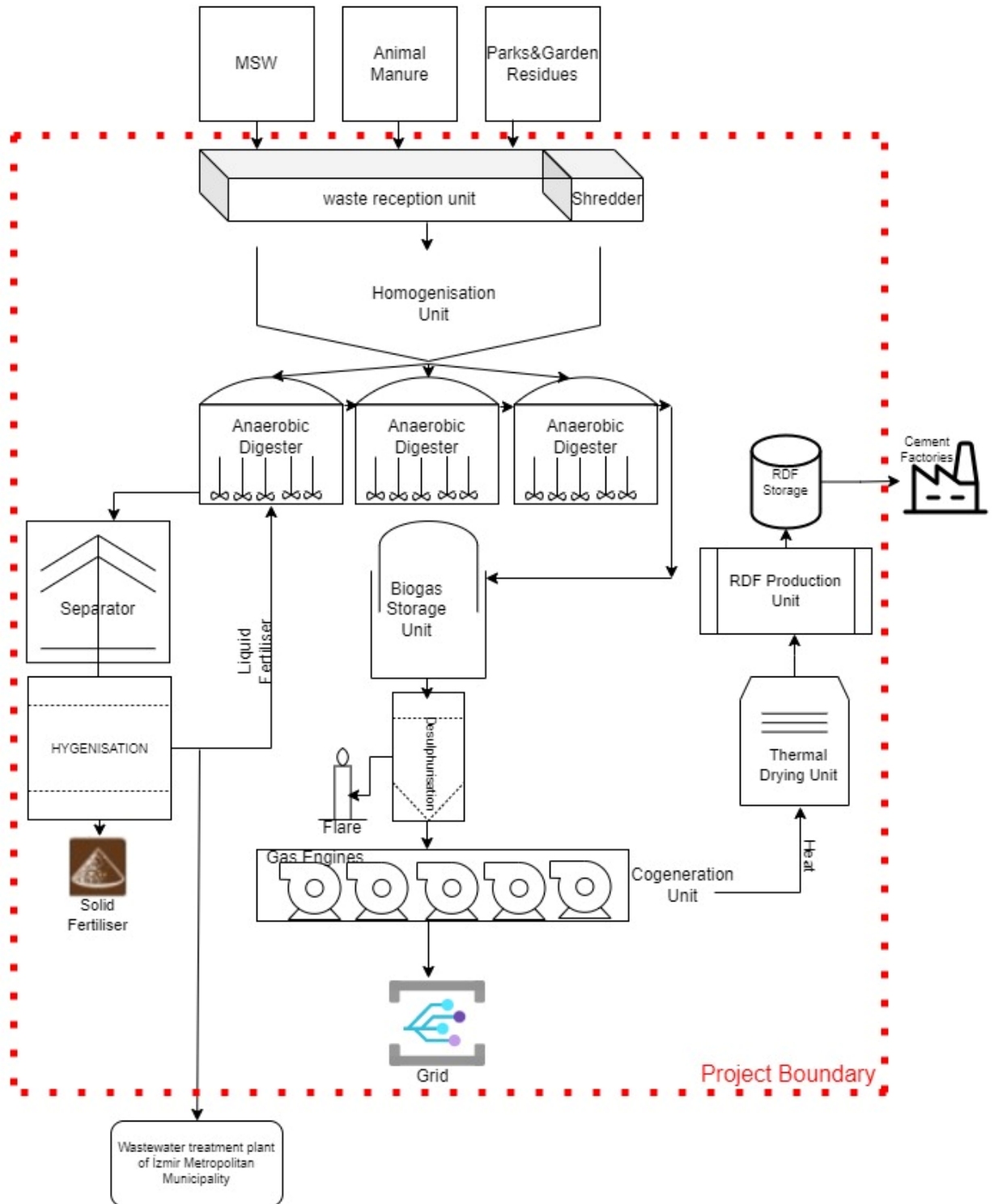


Table 1: The Key Equipment employed in Ödemis ISWMP

Parameters	Number of Units	Capacity & Features	Manufacturer
Anaerobic Digester Tank	3	400 tons/day 20 days retention time 380°C operation temperature	Strabag
Biogas Storage Tank	1	8,500 m ³ storage volume	
Gas Engine	5	1.560 MWe 42.1% electrical efficiency 42.1% thermal efficiency	Marelli MJB 500 MC4
Flare Unit	1	6,000 m ³ /hr 800-900°C combustion temperature 4m lenght	Progeco

The project site is assessed for each kind of environmental sensitivity and is deemed to be suitable for the proposed activities, as stated in the EIA report.

1.12 Project Location

The Project is located on parcels 70 and 71 in the Örnekçiftliği neighbourhood in the district of Ödemiş, Province of İzmir in Turkey.



Figure 1: Satellite view of the project area

KÖŞE KOORDİNATLARI		Koor. Sırası : Sağa, Yukarı		Koor. Sırası : Enlem, Boylam	
		Datum : ED-50		Datum : WGS-84	
		Türü : UTM		Türü : COĞRAFİ	
		D.O.M. : 27		D.O.M. : 27	
		Zon : 35		Zon : 35	
		Ölçek Faktörü : 6 derecelik		Ölçek Fak. : -----	
No	Y	X	Enlem	Boylam	
70 Nolu Parsel	1	584127.1938	4228438.6045	38.1983	27.9603
	2	584098.9701	4228626.9647	38.2000	27.9600
	3	584169.0195	4228627.6908	38.2000	27.9608
	4	584178.9266	4228516.8214	38.1990	27.9609
	5	584205.1955	4228517.0939	38.1990	27.9612
	6	584415.1160	4228541.4689	38.1992	27.9636
	7	584415.8084	4228474.8928	38.1986	27.9636
71 Nolu Parsel	1	584425.3727	4228397.3117	38.1979	27.9637
	2	584136.7554	4228361.0231	38.1976	27.9604
	3	584127.1938	4228438.6045	38.1983	27.9603
	4	584415.8084	4228474.8928	38.1986	27.9636

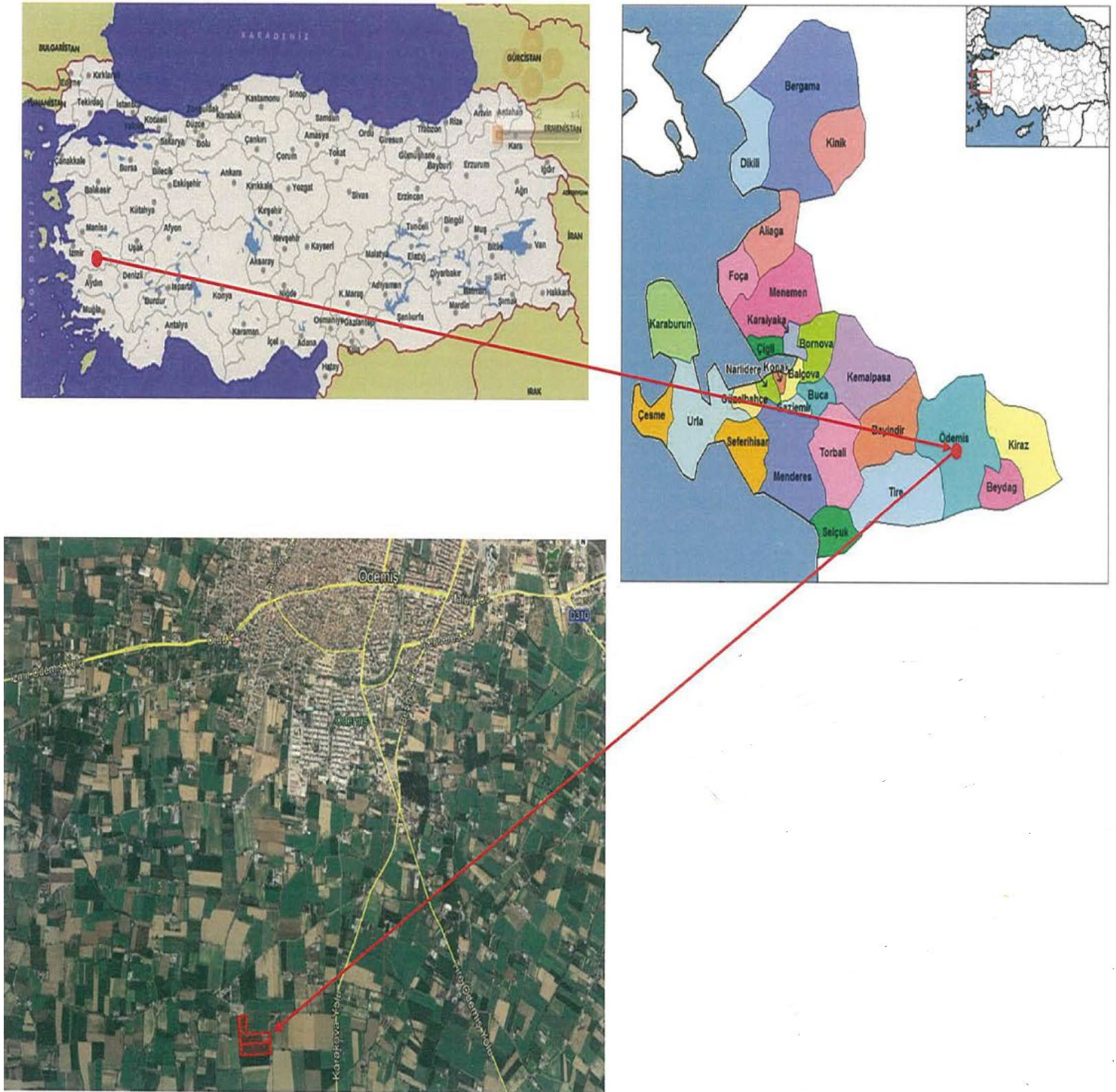


Figure 2: Project Location

1.13 Conditions Prior to Project Initiation

The alternative scenarios of fresh waste in the absence and prior to the project activity include:

- The fresh waste would be disposed of in an SWDS with a partial capture of the LFG and flaring of the captured LFG;
- Electricity would be generated in existing grid-connected electricity plants
- The animal manure would be treated in the uncovered anaerobic lagoons

Before the project activity, the municipal solid waste, agricultural and forestry waste were collected and disposed of in the municipal landfill sites, which do not have any measurements to collect and destroy the landfill gas. The generated wastes from the seven districts were disposed of in two landfill sites. The two municipal landfill sites were located in Harmandalı and Bergama Districts in İzmir.

Before the project activity, there wasn't any recycling activity for municipal mixed waste. The proposed Project doesn't account for the emission reduction due to recycling plastic, glass or metals.

Moreover, without the project activity, the animal manure would be treated in the uncovered anaerobic lagoons, the electricity generated by the Project would be replaced by the power plants connected to the national grid, and fossil fuel fuel-fired boilers would supply the thermal energy.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

Turkish legislation on solid waste management concentrates on the control of solid wastes, packaging wastes, hazardous wastes, medical wastes, and construction wastes. Since municipalities have key roles and responsibilities in waste management services, Municipal Law is also considered in this context.

The related legislation is presented below:

Relevant Law and Regulations for the Project Activity	Comment
Metropolitan Municipalities Law (No: 5216, 10 July 2004)	According to the law, the Metropolitan Municipalities are responsible for preparation of waste management plans, providing, or ensure the provision of, services related to recycling, storage and disposal of municipal, medical, construction and demolition waste. The sub-municipalities within Metropolitan Municipalities are responsible for collection of

	waste at the source, and transferring it to the place of disposal (transfer station) only. The law doesn't mandate any alternative treatment processes for the solid waste management.
Municipal Law (No: 5393, 03 July 2005)	Concerning solid waste, the law states that municipalities are responsible for providing services related to the collection, transfer, sorting, recycling, storage and disposal of waste. The law allows municipalities to perform these services using their own means, i.e., through municipal departments or specialised service units, as well as through a private service provider. Changes in the Law on Metropolitan Municipalities have affected the division of responsibilities. The metropolitan municipality – in provinces where it exists – is now responsible for disposal of solid waste The law doesn't mandate any alternative treatment processes for the solid waste management.
"Environmental Law ² " dated 26/04/2006 numbered 2872 and	The law prohibits the discharge, storage, transfer, and disposal of any kind of waste that does not meet the requirements according to Turkish regulations and standards. It includes the "polluter pays principle" and ensures that the producers take necessary measures to minimise the waste. The Metropolitan Municipalities and the Municipalities are responsible to establish, or have established, and operate landfills. The beneficiaries of waste management services are responsible to contribute the investment, operation, maintenance, and rehabilitation costs via tariffs determined by the Municipal Councils. The law doesn't mandate any alternative treatment processes for the solid waste management.
"Regulation on landfilling ³ " dated 26/03/2010 and numbered 27533	The regulation covers the technical guidelines regarding the landfill facilities, and the procedures and guidelines for the acceptance and storage of the wastes. The regulation also includes provisions on construction of landfill facilities, waste acceptance process, operation of landfill facilities and closing

² See: <https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=2872&MevzuatTur=1&MevzuatTertip=5>

³ See: <http://www.mevzuat.gov.tr/Metin.Aspx?MevzuatKod=7.5.13887&MevzuatIliski=0&sourceXmlSearch=>

	with the control and maintenance period after the closure and rehabilitation of existing facilities. The By-law on Landfill of Waste states that the reduction targets⁴ for the amount of biodegradable municipal waste going to landfills are: 75% by 2015 compared to the production in 2005 50% by 2018 compared to the production in 2005 35% by 2025 compared to the production in 2005. Further, By-law on Landfill of Waste requires the Ministry of Environment and Forestry (now Ministry of Environment and Urbanization) to prepare a National Strategy for the reduction biodegradable municipal waste. The Strategy includes measures of recycling, composting and biogas production or energy/material recovery. The regulation includes measures of recycling, composting and biogas production or energy/material recovery but doesn't mandate any of these alternatives.
Regulation on Incineration of Waste dated 06/10/2010 numbered 27721	The purpose of this By-Law is to prevent and limit the adverse effects of waste incineration on the environment, in particular the pollution of air, soil, surface waters and groundwater as a result of emissions and the risks that may arise for human health by applicable methods. The regulation includes measures of waste incineration but doesn't mandate this alternative process.
"Regulation on Solid Waste Management ⁵ " dated 02/04/2015 and numbered 29314	The regulation aims to determine general waste management principles to protect the environment and human health from generation to disposal of waste. The regulation doesn't mandate any alternative treatment processes for the solid waste management.
Zero Waste Regulation numbered 30829 dated 12/07/2019	The Regulation obliges buildings and campuses that will establish a zero waste management system to comply with the general principles specified in the regulation, to separate and

⁴ Given in the Transitory Article of the Landfill Regulation (O.J No: 27533, 2010/3)

⁵ See: https://webdosya.csb.gov.tr/db/cygm/editordosya/Atik_Yonetimi_Yonetmeligi-2017.docx

	collect waste according to their types, to prevent/reduce waste generation, to comply with the implementation schedule specified in the annexes of the regulation for the establishment and implementation of the zero waste management system, to provide the necessary information trainings on the zero waste management system, to register to the "zero waste information system" established by the Ministry of Environment and Urbanization and to record the requested information and documents in the system. The regulation doesn't mandate any alternative treatment processes for the solid waste management.
Regulation On General Principles Of Waste Pre-Treatment And Recycling Facilities numbered 31623 dated 09/10/2021	The regulation aims that waste is recovered in order to contribute to the economy and reduce the amount of waste to be disposed of in landfills. The regulation includes measures of waste pre-treatment and recycling but doesn't mandate this alternative processes.
Electricity Market Law ⁶ dated 20/02/2001 and numbered 4628	The Law aims to ensure the development of a financially sound and transparent electricity market operating in a competitive environment under civil law provisions and the delivery of sufficient, good quality, low cost and environment-friendly electricity to consumers and to ensure the autonomous regulation and supervision of this market. The regulation doesn't mandate any alternative treatment processes for electricity production.
Law on Utilisation of Renewable Energy Resources for the Purpose of Generating Electricity Energy ⁷ dated 10/05/2005 and numbered 5346	The purpose of this Law is to expand the utilisation of renewable energy resources for generating electrical energy, to benefit from these resources in a secure, economical and qualified manner, to increase the diversification of energy resources, to reduce greenhouse gas emissions, to assess waste products, to protect the environment and to develop the related manufacturing sector for realising these objectives. The Law brings an incentive of 13.3 \$ cent/kWh for the electricity production from biomass. It also brings incentives for local equipment purchases such as turbines, engines, cogeneration systems etc.

⁶ See: <https://www.mevzuat.gov.tr/MevzuatMetin/1.5.4628-20080709.pdf>

⁷ See: <https://www.mevzuat.gov.tr/MevzuatMetin/1.5.5346.pdf>

	The regulation doesn't mandate any alternative treatment processes for renewable electricity production.
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1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The Project has neither been registered nor seeking registration under any other GHG programs. The Project is seeking registration only in VCS program.

1.15.2 Projects Rejected by Other GHG Programs

No, the project activity didn't apply for registration in any other GHG program and was not rejected by any other GHG programme.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the Project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes ☒ No

There are no emissions trading programs or any other mechanism in Turkey that includes GHG allowance trading in Turkey; subsequently, the Project does not fall under any such mechanisms.

1.16.2 Other Forms of Environmental Credit

Has the Project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

The Project has not sought or received any other form of GHG-related environmental credits.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

SDG 13 Climate Action

The Project Activity results in emissions reduction by capturing the methane in the municipal solid waste and converting it to electricity and heat in the cogeneration plant. Without the Project Activity, the municipal solid waste generated would be left to decay in a landfill, producing landfill gas with a partial capture of the LFG and flaring the captured LFG. By avoiding methane emissions, the project activity will result in an average emission reduction of 139,044 tCO₂e annually. Additionally, in the cogeneration plant, captured methane is converted to electricity and heat, where the generated electricity is supplied to the grid, and the generated heat is used for heating the buildings and in the thermal drying unit to dewater the solid fertiliser obtained. Without the Project Activity, the electricity generated would be supplied from the national grid where thermal power plants are connected, causing more GHG emissions. The heat necessary for the building and drying purposes would be provided by the fossil fuels used, causing further GHG emissions.

Additionally, the ISWMP accepts animal manure from nearby farms. Without the Project Activity, the animal manure generated in the farms would be left to decay in an anaerobic lagoon producing methane gas. With the Project Activity, the collected animal manure is being used in the digesters to produce biogas and energy, avoiding the release of methane into the atmosphere along with several other toxic gases.

Moreover, the non-recyclable and combustible part of the incoming waste will be revalued in the ISWMP as refuse-derived fuel and sent to the cement factories to replace the fossil fuel usage, thus preventing the GHG emissions resulting from fossil fuel combustion.

SDG 7 Affordable and Clean Energy

The Project Activity results in renewable energy generation by digesting municipal solid waste and animal manure in an anaerobic environment to produce electricity and heat. The baseline scenario for using gas generated from an anaerobic digester (biogas) is that the electricity delivered by the Project to the National Grid would have otherwise been generated by the operation of grid-connected power plants and/or by the addition of new generation sources into the grid. Turkish electricity generation is mainly composed of thermal power plants, and the share of fossil fuel resources in increasing energy demand of the country; has also increased significantly in recent years. In the absence of the proposed project activity, the same amount of electricity is required to be supplied via either the existing power plants or by increasing the number of thermal power plants, thus increasing GHG emissions.

The clean energy generated by the Project Activity is calculated based on the electricity generated per annum. The Project is expected to generate 32,307.6 MWh of clean energy annually and contributes to the share of renewable energy sources.

In order to estimate the outcome of SDG 7, Indicator 7.2.1, and “Renewable energy share in the total final energy consumption” of SDG Target 7.2: “By 2030, increase the share of renewable energy in the global energy mix substantially” can be employed. The project activity will increase renewable energy share in the country’s energy mix.

There is no electricity generation in the baseline scenario in which Ödemis ISWMP is not being implemented.

The Project’s contribution to SDG 7 is determined via a simple calculation that provides the overall renewable energy generation fed into the grid. This approach is based on the following calculation:

The quantity of net electricity generation supplied by the project plant to the grid in the related year has been calculated by subtracting “Electricity consumption from the grid (MWh)” from “Electricity supplied to the grid (MWh)”

Calculation of Electricity Generation of Ödemis ISWMP:

Net Electricity Generation of Ödemis ISWMP = Electricity supplied to the grid (MWh) - Electricity consumption from the grid (MWh)

SDG 8 Decent Work and Economic Growth

Turkey has a high rate of unemployment. The Project Activity employs more than 200 people, more than half local, and all employee working in the Plant is trained at least once a year. Without this Project, there would be no employment and no training. Therefore, it would not be possible to contribute to the project region's economy and the people's education without the project activity. Health and safety (HSE) training certificates for all employees are provided, and all employees are to be trained according to the requirements. The Project Activity promotes inclusive and sustainable economic growth, employment and decent work for all and contributes to the decrease in the unemployment rate.

SDG 9 Industries, Innovation and Infrastructure

The Project Activity includes the most advanced, fully automated Material Recovery Facility, where state-of-the-art equipment is used to sort out the mixed waste. In fully automated plants sorting efficiency rates are higher than in manual ones. Automated machinery can easily sort some materials which are hard for humans to recognise, and the performance of the machinery is constant for the whole shift. The sorting line equipment consists of a primary shredder, trommel screen, magnets, belt conveyor, eddy current separator, ballistic separator, optic separators, baling press and PLC control system. Using a variety of selection equipment, the Plant can get organics, plastic, metals, brick, and stone out of the mixed municipal solid waste with high efficiency. This Material Recovery Facility has the machinery to transform the collected plastic into value-added plastic waste to be supplied to the recycling companies. Also, anaerobic digesters, biogas storage tanks and dewatering units employed in the biogas plant have state-of-the-art equipment capturing the biogas

effectively and producing a higher quality of fertilisers. State-of-the-art digester presents co-digestion opportunities to manage manure, food waste, and other organics.

Moreover, Refuse-derived fuel (RDF) is produced by processing the mixed municipal solid waste obtained after sorting processes and is used as an alternative to fossil fuels in cement factories. RDF production is a complicated process that requires expertise. It is important to analyse the calorific value of the waste, moisture content, combustion temperature and resulting emissions. For this reason, Ödemis ISWMP has a high-quality waste laboratory.

For all of the above reasons, the Project Activity helps the solid waste industry to innovate new technologies and find more effective solutions to tackle the problem of resource efficiency and increasing waste amounts.

SDG 12 Responsible Consumption and Production

With between 34-40 per cent of the food produced in Turkey going uneaten annually and a current landfill rate of over 76 per cent, our growing mountain of food waste presents an enormous opportunity for businesses looking to do their part for the environment by reducing food waste in the landfill and creating renewable clean energy

Ödemis ISWMP is a closed-loop system meaning all outputs can be used on the Plant itself. Liquid digestate resulting from the digesters is recycled back into the process, reducing the need for adding extra water and often even removing it completely.

The proportion of organics in mixed municipal waste is high in Turkey, from 30 to 60 per cent. Effectively separating these fractions is an essential step in achieving a circular economy. This organic content is then turned into energy in biogas plants. This also reduces the amount of land used for landfilling, and especially the emissions of landfill gases.

The Project Activity will further help Turkey to realise the Zero Waste Project, which was initiated in 2017. Zero Waste System is a seven-step roadmap developed by the Ministry of Environment and Urban Planning, consisting of steps that companies, institutions or organisations should apply to be included in the Zero Waste Project. Between 2017 and 2020, the Project helped save 397 million tons of raw material, 315 million kilowatt-hours of energy, 345 million cubic meters of water and 50 million barrels of oil. In the same period, 17 million tons of usable waste were collected, 2 billion tons of greenhouse gas emissions were prevented, and the Project saved 209 million trees. The economic savings attained with these actions are calculated at 17 billion liras (over 3 billion dollars). The Project aims for direct employment for 100,000 people and 20 billion liras of annual savings⁸.

1.17.2 Sustainable Development Contributions Activity Monitoring

⁸ <http://zerowaste.gov.tr/>

SDG13: Climate Action

The emission reduction parameter can analyse this contribution. The Project's annual average emissions reduction is estimated to be 139,044 tCO₂e for this crediting period.

SDG8: Decent Work and Economic Growth

The Project has employed 219 workers. The employment of local people with the necessary technical qualifications for the required post has been the priority and enhanced by all project activities during the construction and operation of the Ödemis ISWMP. As a result, increased job opportunities and project business activities have partially eliminated local poverty and unemployment. Construction materials for the foundations, cables, and other auxiliary equipment have been preferentially sourced locally. Moreover, as a contribution of the Project to the region's welfare, the quality of the Electricity consumed in the area has been increased by local electricity production, contributing to decreasing distribution losses.

On the other hand, all employees at the facility earn more than the minimum wage. Also, project activity improves the quality of employment by training the employee. Thus, a series of training will be organised with different focuses such as health and safety, environmental management and technical capacity building within the ISWMP.

SDG7: Renewable Affordable and Clean Energy

The Project's gross annual renewable electricity generation has been estimated to be 32,307.6 MWh during this Crediting Period.

The baseline scenario of the project activity is where municipal solid waste is left to decay in landfills with partial capture of landfill gas and flaring of the captured landfill gas. The baseline scenario for the use of gas generated from a biogas power plant is that the Electricity delivered by the Project to the National Grid would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid. Turkish electricity generation is mainly composed of thermal power plants, and the share of renewable resources⁹; has decreased significantly in recent years. There is an increasing demand for Electricity, which is expected to continue in the foreseeable future. In the absence of the proposed project activity, the same amount of Electricity is required to be supplied via either the existing power plants or by increasing the number of thermal power plants, thus increasing GHG emissions.

SDG 9 Industries, Innovation and Infrastructure

⁹ Please see <https://webapi.teias.gov.tr/file/1bfb54a8-26c8-4e65-b354-f47044a1bf6b?download>

The Plant includes a state-of-the-art material recovery facility, anaerobic digesters, biogas storage areas, desulphurisation and dewatering units for fertiliser production, and an RDF Plant. The combination of all these units and equipment is unique to the project activity. Material Recovery Facility is fully automated with no manual interferences. Most of the equipment are constructed in the home country contributing to the economy and the technology. A wastewater treatment plant for the liquid fertiliser will be included in the later stages of the project activity.

SDG 12 Responsible Consumption and Production

The contribution to SDG 12 can be monitored by the number of recovered materials and fertilizer produced by the Project Activity. How much organic waste is consumed in the digesters is also a good indicator of SDG 12 contributions.

The amount of recovered recyclable and combustibles will be monitored during the monitoring periods. The project activity prevents these valuable materials from being disposed of in landfills. In the Capacity Report of the Project Activity, it is estimated that by 2050, the plant will recover 3,832 tonnes of glass, 659 tonnes of paper and cardboard, 12,753 tonnes of plastics and 1,137 tonnes of metals and produce 137,700 tonnes of solid fertiliser and 84,838 tonnes of RDF.

Table 3: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
<i>Sequential row number</i>	<i>SDG Target number</i>	<i>Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator</i>	<i>Indicate the Project's contribution to the SDG Indicator (implemented activities to increase or decrease)</i>	<i>Brief description of the quantifiable impact of the Project's activities related to the SDG indicator, during the monitoring period.</i>	<i>Brief description of the cumulative quantifiable impact of the Project's activities related to the SDG indicator, over the project lifetime.</i>
1)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	the annual average emission reduction of the Project is estimated to be 139,044 tCO ₂ e	the annual average emission reduction of the Project is estimated to be 139,044 tCO ₂ e
2)	8.5	8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities	Implemented activities to increase	During the crediting period, the Project employs a total of 219 workers, with a minimum net monthly salary of 5,500 liras ¹⁰⁾	Since the operation start date of the Plant, the Project has employed a total of 219 workers, with a minimum net monthly salary of 5,500 liras

¹⁰ <https://www.csgeb.gov.tr/asgari-ucret/>

3)	8.8	8.8.2 Level of national compliance with labour rights (freedom of association and collective bargaining) based on International Labour Organization (ILO) textual sources and national legislation, by sex and migrant status	Implemented activities to increase	During the crediting period, the Project provides a total of 219 training to all employee	Since the operation start date of the Project, the Project has provided a total of 219 training to all employee
4)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	During the crediting period, the Project generated a gross of 32,307.6 MWh of renewable electricity	Since the operation start date of the Project, the Project generated a gross of 32,307.6 MWh of renewable electricity
5)	12.3	12.3.1 (a) Food loss index and (b) food waste index	Implemented activities to increase	During the crediting period, the Project has recycled 167,890 tons of food waste into the useful energy	Since the operation start date of the Project, the Project has recycled 167,890 tons of food waste into the useful energy
6)	9.4	Upgrade infrastructure and retrofit industries to make them sustainable, with increased resource-use efficiency and greater adoption of clean and environmentally sound technologies and industrial processes	Implemented activities to increase	During the monitoring period, the state-of-the-art equipment employed in the Plant includes a material recovery facility, anaerobic digesters, biogas storage areas, desulphurisation and dewatering units for fertiliser production, and an RDF Plant.	Since the operation start date of the Project, the state-of-the-art equipment employed in the Plant includes a material recovery facility, anaerobic digesters, biogas storage areas, desulphurisation and dewatering units for fertiliser production, and an RDF Plant. A wastewater treatment plant for the liquid fertiliser will be included in the later stages of the project activity.

1.18 Additional Information Relevant to the Project

Leakage Management

The floor of the waste reception unit is made of 30 cm thick, C30 concrete and non-flammable material to ensure its impermeability. A separate collection mechanism is constructed at the base of the waste reception unit to prevent the waste from contacting or leaking with groundwater, sewage or surface water. The ground is sloped appropriately to collect the leachate in the waste reception area. Also, an impervious concrete floor is applied for leachate on the Material Recovery Facility floor, and grate channels are used to collect the leachate. Moreover, a leachate storage tank is employed in the Plant to transfer the collected leachate to the wastewater treatment plant of İzmir Metropolitan Municipality.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

There is no further information that may have a bearing on the Project's eligibility, the net GHG emission reductions or removals, or the quantification of the Project's net GHG emission reductions or removals.

2 SAFEGUARDS

2.1 No Net Harm

The Ministry of Environment and Urbanization approved the EIA Report for the Project on 12/01/2018. A very detailed and thorough study was conducted for the EIA process of Project Activity. Every aspect of environmental impact has been considered in the EIA report with corresponding measures during project development, and no net harm has been identified. Meanwhile, implementing the Project will improve the environmental conditions of the project region and local socio-economic development by creating employment opportunities.

2.2 Local Stakeholder Consultation

A stakeholder meeting was conducted on 14/07/2020 for Ödemiş ISWMP Project. The meeting location was chosen in the district of Ödemiş, which is the closest settlement to the project area. The meeting was held in the village coffee house, which the local people could easily access. The announcement of the stakeholder meeting was sent by post on 01/07/2020 to the mukhtars of the nearest villages to the project area and also to the Izmir Provincial Directorate of Environment and Urbanization, Izmir Metropolitan Municipality. Subsequently, village mukhtars were also contacted by telephone to provide further information about the meeting. There was participation in the meeting from the local residents, Izmir Provincial Directorate of Environment and Urbanization, Izmir Metropolitan Municipality, the Mayor of Ödemiş Municipality, employees of the Plant, the Project Proponent and the mukhtars of the villages. During the Stakeholder Consultation Meeting, the participants were given detailed information about the planned activity through a visual presentation.

The mechanism for ongoing communication was established with the support of the Mukhtar for stakeholders to forward their requests or concerns regarding the project activities. A logbook for receiving comments from the locals was left in Mukhtar's office. In Turkey, mukhtars are the most relevant persons in the villages for establishing a grievance mechanism. Their offices are some of the few places the locals visit very often for different purposes, such as requesting some official letters. Whenever a problem arises in the village, as the first thing locals go to the mukhtars office to express their concerns to find solutions and reflect on their issues with the Mukhtar. It has been decided that the Mukhtar will also take an active role in monitoring the participants' requests so that a grievance mechanism situating the Mukhtar at the centre was established for this project activity. Mukhtar was provided with all the contact information of the responsible persons for the project activity. Whenever a complaint arises from the local people or Mukhtar himself is concerned about the project activity, he can directly contact the relevant responsible persons.

In general, local stakeholders were supportive of the project activity. Most of the local stakeholders thought the Project would help improve the livelihood of local people and promote local economic development without much adverse environmental impact. Therefore, the project implementation is regarded as beneficial by most of the stakeholders. No negative comments from the stakeholders were received during this meeting.

2.3 Environmental Impact

Within the scope of the environmental impact assessment study carried out for the project activity, the following key environmental impacts were assessed.

Air Pollution

A detailed study has been carried out within the scope of EIA regarding the emissions expected to occur during the construction and operation periods of the project activity. Electricity is used as an energy source in Plant units, and there are emissions from waste transportation activities to the material recovery facility and biogas plant. These emission amounts are calculated in the scope of EIA. Based on the resulting calculations, it was determined that the emissions to be generated are below the regulatory limit values. Biogas filters and biogas treatment units are employed in the biogas plant to ensure that pollutant emissions in the plant are kept well below the regulatory thresholds.

Water Pollution

Most liquid fertiliser obtained from the biogas plant is recycled back into the system. The remaining liquid fertiliser product formed in the Biogas Plant is supplied to the market as liquid fertiliser after obtaining the necessary permits within the scope of the "Regulation on Organic, Mineral and Microbial Source Fertilizers Used in Agriculture". However, suppose the liquid fertiliser product does not meet the fertiliser criteria. In that case, the pollutant levels will be reduced to the desired levels in the pre-treatment unit constructed in the ISWMP and then sent to the wastewater treatment plant under the responsibility of Izmir Metropolitan Municipality of Turkey by sewage trucks. At every stage of the Project, the provisions of the "Water Pollution Control Regulation", which entered into force after being published in the Official Gazette dated 31.12.2004 and numbered 25687, will be complied with.

Soil Pollution and Solid Waste

The Project aims to minimise the adverse effects of inadequate waste management practices and use solid wastes as a resource. Biodegradable, recyclable, and residual wastes are obtained from the Material Recovery Facility. Biodegradable wastes are sent to the Biogas Plant, and recyclable wastes are then pressed and shipped to licensed recovery/recycling companies. The remaining waste that cannot be recovered is sent to the landfill. Recyclable wastes generated during the construction and operation phases of the

Project are collected separately and delivered to the company at certain intervals within the framework of the "Regulation on the Control of Packaging Wastes", dated 27.12.2017 and numbered 30283. In accordance with Article 18 of the "Regulation on Control of Waste Oils", waste oils are collected in tanks/containers placed on the impermeable ground, different tanks/containers are used for different categories of waste oils, and waste oil temporary storage tanks/containers are equipped with indicators and overfill prevention devices.

Hazardous wastes generated due to the Project Activity are segregated by type and are stored in a sealed, covered temporary waste storage. There are signboards in each hazardous waste storage tank containing information (name, waste code, etc.) to identify the waste. During the operation phase of the Project, waste batteries are collected in waste battery containers placed at certain points within the facility and delivered to the "Association of Portable Battery Manufacturers and Importers" as soon as these waste battery containers are full

The ISWMP and its units are constructed on impermeable reinforced concrete ground. Excavation wastes generated during the project area's construction activities were removed from the facility, transferred to the landfill site and not left on the plant floor.

Noise Pollution

The machinery and equipment used during the Project's operation phase operate in a sequence and are unlikely to operate simultaneously. However, the noise calculations conducted within the scope of the EIA are based on the most unfavourable scenario. It is assumed that all of the equipment operate simultaneously. When the noise levels given in the Acoustic Report/Table.9. in Annex-14 of EIA are compared with the regulatory noise limit values, it is seen that the limit value of 70 dBA is met after a distance of approximately 125 m during the operation phase of the activity for daytime.

Considering that the closest settlement to the project area is 2 km away from the project area, the settlement will not be affected.

2.4 Public Comments

This Project will be open for public comment on the Verra website after being listed on the pipeline.

2.5 AFOLU-Specific Safeguards

N/A.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The Project Activity applies ACM0022 - Alternative waste treatment processes (version 03.0). In line with this large-scale methodology, the following CDM methodological tools are also applied:

- “Combined tool to identify the baseline scenario and demonstrate additionality” (version 07.0);
- “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0);
- “Emissions from solid waste disposal sites” (version 08.0);
- “Project and leakage emissions from anaerobic digesters” (version 02.0);
- “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” (version 03.0).
- “Project Emissions from Flaring” (version 04.0);

The Project Activity applies ACM0010- “GHG emission reductions from manure management systems” (version 08.0). In line with this large-scale methodology, the following CDM methodological tools are also applied:

- Combined tool to identify the baseline scenario and demonstrate additionality (Version 7.0);

3.2 Applicability of Methodology

Demonstration of applicability conditions for the CDM baseline and monitoring methodology ACM0022 (version 03.0) and all above-referred CDM methodological tools are included in the tables below:

Applicability Condition of ACM0022 – “Alternative waste treatment processes” (version 03.0)	Justification
The methodology applies to project activities that install and operate new plants for the treatment of fresh waste through any combination of the following processes:	The project activity includes the installation and operation of a new facility to treat fresh organic waste through anaerobic digestion with biogas recovery and its utilisation as a gaseous fuel for electricity

(a) Composting process under aerobic conditions; (b) Anaerobic digestion with biogas recovery and flaring and/or its use; (c) Co-composting of wastewater in combination with solid waste; (d) Anaerobic co-treatment of wastewater in combination with solid waste; (e) Mechanical/thermal treatment process to produce refuse-derived fuel (RDF) or stabilised biomass (SB) that is produced within the project boundary and its use; (f) Gasification process to produce syngas and its use; (g) Incineration of fresh waste for the generation of thermal/electric energy. (h) The following conditions apply to all project activities using this methodology: (i) The project plant only treats fresh waste/wastewater for which emission reductions are claimed, except for cases involving composting, co-composting and anaerobic digestion; (j) Neither the fresh waste nor the products from the project plant are stored on-site under anaerobic conditions; (k) Any wastewater discharge resulting from the project activity is treated in accordance with applicable regulations; (l) The project activity does not reduce the amount of waste that would be recycled in the absence of the project activity. This shall be	generation, which meets the applicability condition of (b). Additionally, The project activity encompasses the installation and operation of Mechanical/thermal treatment process to produce refuse-derived fuel (RDF) within the project boundary, which meets the applicability condition of (e). Furthermore, the project activity only treats fresh solid organic waste for which emission reductions are claimed. Thus, the project activity meets condition (i). No fresh solid waste from the project plant is stored on-site under anaerobic conditions. Thus, the project activity meets the requirement (j). The project activity does not involve wastewater discharge, which doesn't conform to the applicable regulations. Thus, the project activity meets condition (k). The project activity does not reduce the amount of waste that would be recycled in the absence of the project activity. Thus, the project activity meets condition (l). There are no regulations mandating any waste treatment process implemented under the project activity. Therefore, condition (m) is not applicable. The project activity does not involve the treatment of hazardous waste. Thus, the project activity meets Condition (n).
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justified and documented in the clean development mechanism project design document (CDM-PDD); (m) When applicable regulations mandate any waste treatment process implemented under the project activity, the rate of compliance with such regulations for the treatment process is below 50 per cent; (n) Hazardous wastes/wastewater are not eligible under this methodology.	
The methodology is only applicable if the baseline scenario is: (a) The disposal of the fresh waste in an SWDS with or without a partial LFG capture system (M2 or M3); (b) In the case of co-composting or co-treatment of wastewater in an anaerobic digester, the treatment of organic wastewater in either an existing or new anaerobic lagoon or sludge pit without methane recovery (W1 or W4); (c) In the case of electricity generation, the electricity is generated in an existing/new captive fossil fuel fired power-only plant, captive cogeneration plant and/or the grid (P2, P4 or P6); (d) In the case of heat generation, the heat is generated in an existing/new fossil fuel fired cogeneration plant, boiler or air heater (H2 or H4).	The baseline scenario for the project activity is the disposal of the fresh waste in an SWDS with a partial LFG capture system as stated in requirement (a) There is no co-composting or co-treatment of wastewater in the Plant's anaerobic digesters. So, condition (b) is not applicable. In the case of electricity generation, without the project activity, the electricity would have been generated by the grid, which primarily consists of thermal power plants meeting the requirement (c). In the case of heat generation, without the project activity, the heat would have been generated by the boilers utilising fossil fuels meeting the requirement (d)

Specific applicability conditions for the different processes are provided in Table 2 below:				In line with Table 2, Waste treatment option under the project activity is anaerobic digestion, thermal treatment and mechanical treatment. The conditions of each waste treatment option are analysed as follows: Anaerobic Digestion: Applicable types of wastes that may be treated are wastewater and fresh waste; for the project activity, fresh waste is being utilised in anaerobic digesters. In line with Table 2, Applicable products and their use, biogas may be flared, used to generate electricity or heat, and/or upgraded and distributed in a natural gas distribution grid; for the project activity, biogas is used to generate electricity and heat. In line with Table 2, Applicable waste by-products, Non-biodegradable materials that may have a market value (i.e. glass, metals and plastics); Wastewater discharge and Digestate; for the project activity digestate is produced in the anaerobic digesters that have a market value. Applicability conditions of “TOOL14: Project and leakage emissions from anaerobic digesters” are assessed below. Thermal Treatment Applicable types of wastes that may be treated are fresh waste; for the project activity, fresh waste is being utilised in Thermal Treatment. In line with Table 2, Applicable products and their use, RDF/SB: any use is applicable; for the project activity, RDF is produced.
Table 2. Specific applicability conditions for the different waste treatment processes				
Waste treatment option under the project activity	Applicable types of wastes that may be treated	Applicable products and their use	Applicable waste by-products	
Composting or co-composting	Types of waste as specified in the scope and applicability section of “TOOL13: Project and leakage emissions from composting”; Wastewater; Wastewater discharge	Compost: any use applicable	Non-biodegradable materials that may have market value (i.e. glass, metals and plastics); Wastewater discharge	
Anaerobic digestion	Wastewater; Fresh waste; Wastewater discharge	Biogas may be flared, used to generate electricity or heat, and/or upgraded and distributed in a natural gas distribution grid	Non-biodegradable materials that may have market value (i.e. glass, metals and plastics); Wastewater discharge; Digestate	
Thermal treatment	Fresh waste	RDF/SB: any use is applicable	Non-biodegradable materials that may have market value (i.e. glass, metals and plastics)	
Mechanical treatment	Fresh waste	RDF/SB: any use is applicable	Wastewater discharge; Non-biodegradable materials that may have market value (i.e. glass, metals and plastics)	
Gasification	Fresh waste	Syngas may be used to generate electricity and/or heat	Gasification by-products (e.g. inert materials); Wastewater discharge; Non-biodegradable materials that may have market value (i.e. glass, metals and plastics)	

	In line with Table 2, Applicable waste by-products, Non-biodegradable materials may have a market value (i.e. glass, metals and plastics); for the project activity, Non-biodegradable materials may have a market value obtained. Mechanical Treatment Applicable types of wastes that may be treated are fresh waste; for the project activity, fresh waste is being utilised in Mechanical Treatment. In line with Table 2, Applicable products and their use, RDF/SB: any use is applicable; for the project activity, RDF is produced. In line with Table 2, Applicable waste by-products, wastewater discharge, and Non-biodegradable materials may have a market value (i.e. glass, metals and plastics); for the project activity, Non-biodegradable materials may have a market value obtained.
Applicability Condition of CDM Tool 14 – “Project and Leakage emissions from anaerobic digesters” (version 02.0)	Justification
This tool provides procedures to calculate project and leakage emissions associated with anaerobic digestion in an anaerobic digester. The tool is not applicable to other systems where waste may be decomposed anaerobically, for instances stockpiles, SWDS or unaerated lagoons.	The project activity includes the treatment of a biodegradable fraction of solid waste through anaerobic digestion in an anaerobic digester. The project activity meets the applicability condition of the methodological tool.
Applicability Condition of CDM Tool 04 – “Emissions from solid waste disposal sites” (version 08.0)	Justification

The tool can be used to determine emissions for the following types of applications: (a) Application A: The CDM project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g. “ACM0001: Flaring or use of landfill gas”). The methane is generated from waste disposed in the past, including prior to the start of the CDM project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (CDM-PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e.g. measuring the amount of methane captured from the SWDS); (b) Application B: The CDM project activity avoids or involves the disposal of waste at a SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW) is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	Application B is relevant for this Project Activity. The project activity avoids the disposal of waste at an SWDS. The solid waste (MSW) is treated with anaerobic digestion and is then prevented from being disposed of in an SWDS. The methane is avoided from disposal during the crediting period. The tool is applied for both ex-ante and ex-post estimation of emissions.
Applicability Condition of CDM Tool 05 – “Baseline, project and/or leakage emissions from electricity consumption	Justification

and monitoring of electricity generation” (version 03.0)	
If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or³⁹ (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	The tool is applied for calculating emissions from electricity consumption and Scenario A is applicable to the sources of electricity consumption. For the Project activity energy consumption is achieved from the grid only and either no captive power plant(s) is/are installed at the site of electricity consumption.

This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	In the Project Activity, electricity is supplied to the grid. The project activity meets Scenario I.
This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO2 emissions.	In the baseline scenario, captive renewable power generation technologies don't provide electricity to the project activity. With the tool, only CO2 emissions are calculated.
Applicability Condition of CDM Tool 06 – “Project emissions from flaring” (version 04.0)	Justification
The tool is applicable to enclosed or open flares	The project activity employs an enclosed flare.
This tool is applicable to the flaring of flammable greenhouse gases where: (a) Methane is the component with the highest concentration in the flammable residual gas; and (b) The source of the residual gas is coal mine methane or a gas from a biogenic source (e.g. biogas, landfill gas or wastewater treatment gas).	For the project activity; (a) Methane is the component with the highest concentration in the residual gas; and (b) The source of the residual gas is biogenic, namely biogas.

The tool is not applicable to the use of auxiliary fuels and therefore the residual gas must have sufficient flammable gas present to sustain combustion. In the case of an enclosed flare, there shall be operating specifications provided by the manufacturer of the flare and these shall be followed by the project participant	The project activity doesn't use auxiliary fuels for the combustion of biogas. The project activity complies with the manufacturer's specifications for the operation of the flare.
Applicability Condition of CDM Tool 08 – “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” (version 03.0)	Justification
Typical applications of this tool are methodologies where the flow and composition of residual or flared gases or exhaust gases are measured for the determination of baseline or project emissions	In the Project Activity, the flow and composition of residual gas are measured for the determination of baseline or project emissions.
Methodologies where CO₂ is the particular and only gas of interest should continue to adopt material balances as the means of flow determination and may not adopt this tool as material balances are the cost effective way of monitoring flow of CO₂	In the Project Activity, CH₄ is the particular and only gas of interest.
The underlying methodology should specify: (a) The gaseous stream the tool should be applied to; (b) For which greenhouse gases the mass flow should be determined; (c) In which time intervals the flow of the gaseous stream should be measured;	In the Project Activity, (a) the gaseous stream is biogas produced in the anaerobic digesters (b) The mass flow of Methane is determined (c) Continuous measurements and hourly recording of the biogas flow are achieved.
Applicability Condition of CDM Tool 02 – “Combined tool to identify the baseline scenario and demonstrate additionality” (version 07.0)	Justification

The tool is applicable to all types of proposed project activities. However, in some cases, methodologies referring to this tool may require adjustments or additional explanations as per the guidance in the respective methodologies.	This tool is applied to define the alternatives to the project activity with adjustments and additional explanations as per the guidance in the ACM0022 – “Alternative waste treatment processes” (version 03.0)
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Applicability Condition of ACM0010 – “GHG emission reductions from manure management systems” (version 08.0)	Justification
This methodology is applicable to manure management on livestock farms where the existing anaerobic manure treatment system, within the project boundary, is replaced by one or a combination of more than one animal waste management systems (AWMSs) that result in less GHG emissions compared to the existing system. The methodology is also applicable to Greenfield facilities.	The existing anaerobic manure treatment systems in nearby farms are replaced by the anaerobic digestion system in Project Activity which results in less GHG emissions.
This methodology is applicable to manure management projects under the following conditions: (a) Farms where livestock populations, comprising of cattle, buffalo, swine, sheep, goats, and/or poultry, is managed under confined conditions; (b) Farms where manure is not discharged into natural water resources (e.g. rivers or estuaries); (c) In case of anaerobic lagoons treatments systems, the depth of the lagoons used for manure management under the baseline scenario should be at least 1 m; (d) The annual average ambient temperature at the site where the anaerobic	In the Project Activity; (a) Livestock populations of the nearby farms are managed under confined conditions. (b) The farms don’t discharge manure into natural water resources in the baseline. (c) The farms have lagoons with a depth of at least 1 m in the baseline. (d) The annual average ambient temperature in the project region is higher than 5°C. (e) The retention time of the animal manure in the anaerobic lagoons of the farms is greater than 1 month in the baseline.

manure treatment facility in the baseline existed is higher than 5°C; (e) In the baseline case, the minimum retention time of manure waste in the anaerobic treatment system is greater than one month; (f) The AWMS(s) in the project case results in no leakage of manure waste into ground water, for example the lagoon should have a non-permeable layer at the lagoon bottom.	(f) Anaerobic Digestion system employed in the project activity results in no leakage of manure waste into groundwater since the lagoons have a non-permeable layer at the bottom.
Applicability Condition of CDM Tool 02 – “Combined tool to identify the baseline scenario and demonstrate additionality” (version 07.0)	Justification
The tool is applicable to all types of proposed project activities. However, in some cases, methodologies referring to this tool may require adjustments or additional explanations as per the guidance in the respective methodologies.	This tool is applied to define the alternatives to the project activity with adjustments and additional explanations as per the guidance in the ACM0010 – “GHG emission reductions from manure management systems” (version 08.0)

3.3 Project Boundary

ACM0022 Alternative Waste Treatment Processes Version 03.0

The spatial extent of the project boundary is the SWDS, where the waste is disposed of in the baseline, anaerobic lagoons or sludge pits treating organic wastewater in the baseline, and the site of the alternative waste treatment process (es). The boundary also includes on-site electricity and/or heat generation and use, on-site fuel use and the wastewater treatment plant used to treat the wastewater by-products of the alternative waste treatment process (es). The project boundary does not include facilities for waste collection and transport

In the case that the Project provides electricity to a grid, then the spatial extent of the project boundary will also include those plants connected to the energy system to which the Plant is connected. If upgraded biogas is fed to a natural gas distribution system, then the natural gas distribution system is also included in the boundary.

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from heat generation	CO ₂	Yes	CO2 emission resulting from heat generation are accounted for.
		CH ₄	No	
		N ₂ O	No	
	Emissions at the SWDS	CO ₂	No	CO2 emissions from the decomposition of fresh waste are not accounted for
		CH ₄	Yes	Major emission source.
		N ₂ O	No	N2O emissions are small compared to CH4 emissions from landfills. The exclusion of this gas is conservative.
	Emissions from electricity generation	CO ₂	Yes	Major source since electricity generation is included in the project activity and is sent to the grid
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
Project	Emissions from on-site fossil fuel consumption due to the project activity other than for electricity generation	CO ₂	No	Excluded for simplification. This emission source is assumed to be very small.
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
	Emissions from on-site electricity use	CO ₂	Yes	May be an important emission source
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.

Source		Gas	Included?	Justification/Explanation
	Emissions from the waste treatment processes	CO ₂	No	Since the project activity doesn't include incineration, gasification or combustion of fossil-based waste, the CO ₂ emissions are excluded. This is conservative.
		CH ₄	Yes	CH ₄ leakage from the anaerobic digester and incomplete combustion in the flaring process are potential sources of project emissions.
		N ₂ O	No	Since the project activity doesn't include RDF combustion, the N ₂ O emissions are excluded. This is conservative.
	Emissions from wastewater treatment	CO ₂	No	Since the project activity doesn't include wastewater treatment, the emissions are excluded. This is conservative
		CH ₄	No	Since the project activity doesn't include wastewater treatment, the emissions are excluded. This is conservative
		N ₂ O	No	Since the project activity doesn't include wastewater treatment, the emissions are excluded. This is conservative

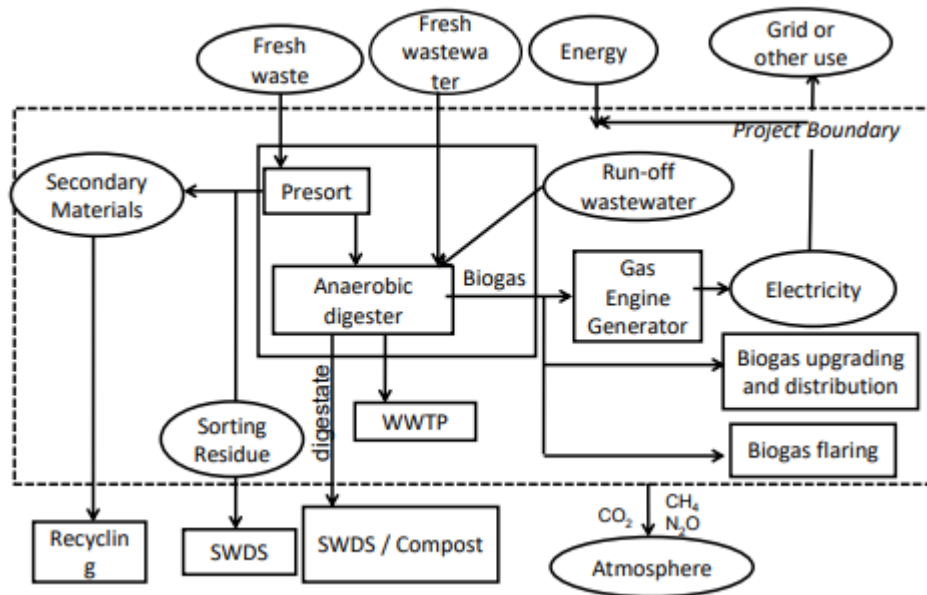
ACM0010 GHG emission reductions from manure management systems Version 08.0

The spatial extent of the project boundary encompasses the site of the AWMS(s), including the flare or energy and/or heat generation equipment and the power/heat source

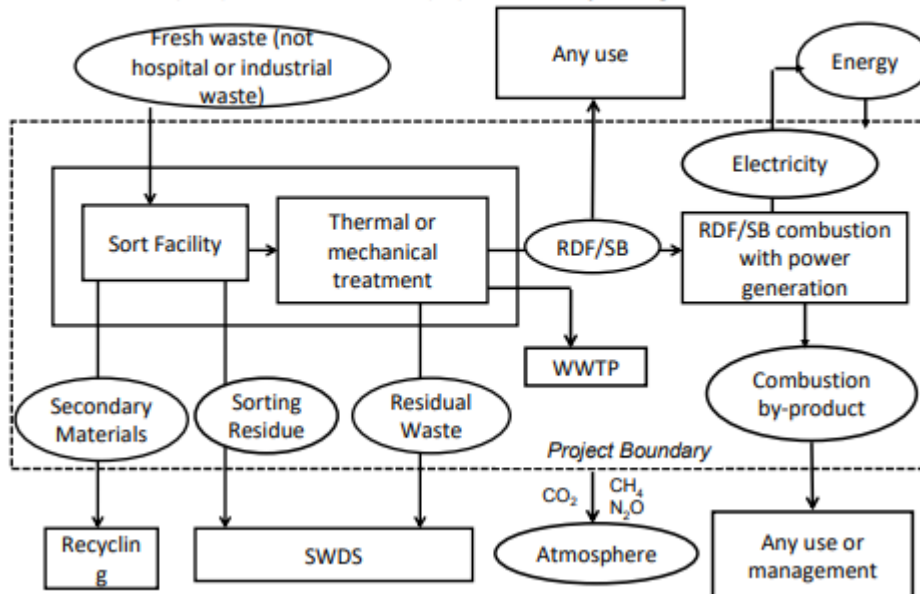
Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from the waste treatment processes	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted
		CH ₄	Yes	The major source of emissions in the baseline
		N ₂ O	Yes	Direct and indirect N ₂ O emissions are accounted
	Emissions from electricity consumption/generation	CO ₂	Yes	Electricity is consumed from the grid in the baseline scenario
		CH ₄	No	Excluded for simplification. This is conservative.

Source		Gas	Included?	Justification/Explanation
Project	Emissions from thermal energy generation	N ₂ O	No	Excluded for simplification. This is conservative.
		CO ₂	Yes	Although thermal energy is generated in the project activity, emissions related to the thermal energy generations are accounted for.
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
	Emissions from on-site electricity use	CO ₂	No	These emissions are accounted for since the project activity consumes the electricity from the grid.
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
	Emissions from the waste treatment processes	CO ₂	No	CO2 emissions from the decomposition of organic waste are not accounted
		CH ₄	Yes	The emission from anaerobic digesters is accounted for.
		N ₂ O	No	Direct and indirect N2O emissions are accounted.

Anaerobic Digestion with biogas collection and flaring and/or its use



Mechanical/thermal treatment process to produce refuse-derived fuel (RDF)/stabilized biomass (SB) and on-site power generation



3.4 Baseline Scenario

In line with the applied methodology ACM0022 (version 03.0), the identification of the baseline scenario and the demonstration of the additionality of the project activity are achieved using the latest version of the CDM methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality” (version 07.0).

The following steps from the above-mentioned methodological tool are followed:

Step 0 – Demonstration whether the proposed project activity is the First-of-its kind;

Step 1 – Identification of alternative scenarios

Step 2 – Barrier analysis

Step 3 – Investment analysis

Step 4 – Common practice analysis

Step 0 – Demonstration whether the proposed project activity is the First-of-its kind;

Since the Project Activity is not the First-of-its kind, Step 0 is not applied.

Step 1 – Identification of alternative scenarios

Step 1a: Define alternative scenarios to the proposed CDM project activity

As per the applied methodology ACM0022 “Alternative waste treatment processes”;

For the treatment of fresh waste, the following alternatives are considered:

M1: The project activity without being registered as a CDM project activity (i.e. any (combination) of the waste treatment processes listed in Table 2);

M2: Disposal of the fresh waste in a SWDS with a partial capture of the LFG and flaring of the captured LFG;

M3: Disposal of the fresh waste in a SWDS without a LFG capture system;

M4: Part of the fresh fraction of the solid waste is recycled and not disposed in the SWDS;

M5: Part of the fresh fraction of the solid waste is treated aerobically and not disposed in the SWDS;

M6: Part of the organic fraction of the solid waste is incinerated and not disposed in the SWDS;

M7: Part of the organic fraction of the solid waste is gasified and not disposed in the SWDS;

M8: Part of the organic fraction of the solid waste is treated in an anaerobic digester and not disposed in the SWDS;

M9: Part of the organic fraction of the solid waste is mechanically or thermally treated to produce RDF/SB and not disposed in the SWDS.

For electricity generation the following alternatives or combinations of these alternatives, inter alia, shall be considered:

P1: Electricity generated as an output of one of the waste treatment processes listed in Table 1, not undertaken as a CDM project activity;

P2: Use of an existing or construction of a new on-site or off-site fossil fuel fired cogeneration plant;

P3: Existing or new construction of an on-site or off-site renewable based cogeneration plant;

P4: Existing or new construction of an on-site or off-site fossil fuel fired electricity plant;

P5: Existing or new construction of an on-site or off-site renewable based electricity plant;

P6: Electricity generation in existing and/or new grid-connected electricity plants.

For heat generation, one or a combination of the following alternatives, inter alia, shall be considered:

H1: Heat generated as a by-product from one of the processes for waste treatment listed in Table 1, not undertaken as a CDM project activity;

H2: Use of (an) existing or construction of (a) new on-site or off-site fossil fuel fired cogeneration plant(s);

H3: Use of (an) existing or construction of (a) new on-site or off-site renewable based cogeneration plant(s);

H4: Use of (an) existing or construction of (a) new on-site or off-site fossil fuel based boiler(s) or air heater(s);

H5: Use of (an) existing or construction of (a) new on-site or off-site renewable energy-based boiler(s) or air heater(s);

H6: District heat;

H7: Other heat generation technologies (e.g. heat pumps or solar energy).

Assessment and demonstration of potential waste management alternatives are shown in table below.

Alternative Scenario		A realistic Alternative?	Justification
M1	The project activity without being registered as a CDM project activity	Yes	This scenario is assessed as a realistic scenario

Alternative Scenario		A realistic Alternative?	Justification
	(i.e. any (combination) of the waste treatment processes listed in Table 2 of ACM0022;		because there are no mandatory laws or regulations to enforce the registration of the project in the CDM mechanism.
M2	Disposal of the fresh waste in an SWDS with a partial capture of the LFG and flaring of the captured LFG;	No	Since Turkey has no regulation to enforce landfill gas capture, this scenario is assessed as non-realistic.
M3	Disposal of the fresh waste in a SWDS without a LFG capture system;	Yes	Since Turkey has no regulation to enforce landfill gas capture, this scenario is considered realistic.
M4	Part of the fresh fraction of the solid waste is recycled and not disposed in the SWDS;	No	Landfilling is the most common practice of waste management in Turkey. The waste is collected in a mixed way and not recycled. So this alternative is assessed as a non-realistic alternative.
M5	Part of the fresh fraction of the solid waste is treated aerobically and not disposed in the SWDS;	No	Aerobic treatment of fresh waste is not a common waste management practice in Turkey. For this reason, this alternative is assessed as a non-realistic alternative.
M6	Part of the organic fraction of the solid waste is incinerated and not disposed in the SWDS;	No	The incineration of fresh waste is not a common waste management practice in Turkey. Also, this alternative would

Alternative Scenario		A realistic Alternative?	Justification
			require serious investments and high know-how. For this reason, this alternative is assessed as a non-realistic alternative
M7	Part of the organic fraction of the solid waste is gasified and not disposed in the SWDS;	No	The gasification of fresh waste is not a common waste management practice in Turkey. Also, this alternative would require serious investments and high know-how. For this reason, this alternative is assessed as a non-realistic alternative
M8	Part of the organic fraction of the solid waste is treated in an anaerobic digester and not disposed in the SWDS;	No	The anaerobic digestion of fresh waste is not a common waste management practice in Turkey. Also, this alternative would require serious investments and high know-how. For this reason, this alternative is assessed as a non-realistic alternative
M9	Part of the organic fraction of the solid waste is mechanically or thermally treated to produce RDF/SB and not disposed in the SWDS.	No	The mechanical or thermal treatment to produce RDF is not a common waste management practice in Turkey. Also, this alternative would require serious investments and high know-how. For this reason, this alternative is

Alternative Scenario		A realistic Alternative?	Justification
			assessed as a non-realistic alternative.
P1	Electricity generated as an output of one of the waste treatment processes listed in Table 2 of the ACM0022 version 03.0, not undertaken as a CDM project activity;	No	Since none of the Waste treatment options listed in Table 2 is common practice for waste management in Turkey; therefore, P1 is unrealistic.
P2	Use of an existing or construction of a new on-site or off-site fossil fuel fired cogeneration plant;	No	Since Project Activity uses and supplies electricity to the national grid, without the project activity same electricity would have been also provided by the grid. This alternative is unrealistic.
P3	Existing or new construction of an on-site or off-site renewable based cogeneration plant	No	Since renewable based cogeneration plant is not common practice for energy generation in Turkey, this alternative is unrealistic.
P4	Existing or new construction of an on-site or off-site fossil fuel fired electricity plant	No	The Project Activity supplies its electricity from the national grid. So this alternative is unrealistic.
P5	Existing or new construction of an on-site or off-site renewable based electricity plant;	No	The Project Activity supplies its electricity from the national grid. So this alternative is unrealistic.
P6	Electricity generation in existing and/or new grid-connected electricity plants.	Yes	Electricity generated by the project activity would have been generated and exported through the

Alternative Scenario		A realistic Alternative?	Justification
			national grid without the project activity. Current grid-connected thermal power plants would have generated the same electricity. This alternative is assessed as a realistic alternative.
H1	Heat generated as a by-product from one of the processes for waste treatment listed in Table 1, not undertaken as a CDM project activity	No	Since none of the Waste treatment options listed in Table 2 is common practice for waste management in Turkey; heat generation by one of these treatment option is unrealistic. Therefore, P1 is unrealistic.
H2	Use of (an) existing or construction of (a) new on-site or off-site fossil fuel fired cogeneration plant(s)	No	Since cogeneration plants are not common in Turkey and require more investments and know-how and also there are no legal enforcement in Turkey for cogeneration plants, this alternative is not realistic.
H3	Use of (an) existing or construction of (a) new on-site or off-site renewable based cogeneration plant(s)	No	Since renewable-based cogeneration plants are not common in Turkey, this alternative is not realistic.
H4	Use of (an) existing or construction of (a) new on-site or off-site fossil fuel based boiler(s) or air heater(s)	Yes	Without the Project Activity, the fossil fuel-based boilers would have supplied the necessary heat. This is the most common practice of heat generation in Turkey. This alternative is realistic.

Alternative Scenario		A realistic Alternative?	Justification
H5	Use of (an) existing or construction of (a) new on-site or off-site renewable energy-based boiler(s) or air heater(s)	No	Since renewable energy-based boiler(s) or air heater(s) are not standard practices in Turkey and there are no legal enforcements, this alternative is unrealistic.
H6	District heat	No	District heating is not standard in Turkey and is also associated with high costs, and this alternative is unrealistic.
H7	Other heat generation technologies (e.g. heat pumps or solar energy)	No	Heat generation by heat pumps or solar energy is not common in Turkey. These technologies require know-how and high investments, and no legal enforcement exists. So this alternative is not realistic.

Plausible Alternative Scenarios identified are given in the table below:

Alternative Scenario		A realistic Alternative?	Justification
M1	The project activity without being registered as a CDM project activity (i.e. any (combination) of the waste treatment processes listed in Table 2 of ACM0022;	Yes	This scenario is assessed as a realistic scenario because there are no mandatory laws or regulations to enforce the registration of the project in the CDM mechanism.
M3	Disposal of the fresh waste in a SWDS without a LFG capture system;	Yes	Since Turkey has no regulation to enforce landfill gas capture, this

			scenario is considered realistic.
P6	Electricity generation in existing and/or new grid-connected electricity plants.	Yes	No electricity is generated and exported through the national grid without the project activity. Current grid-connected thermal power plants would have generated the same electricity. This alternative is assessed as a realistic alternative.
H4	Use of (an) existing or construction of (a) new on-site or off-site fossil fuel based boiler(s) or air heater(s)	Yes	Without the Project Activity, the fossil fuel-based boilers would have supplied the necessary heat. This is the most common practice of heat generation in Turkey. This alternative is realistic.

In line with the applied methodology ACM0010 (version 08.0), the identification of the baseline scenario and the demonstration of the additionality of the project activity are achieved using the “Combined tool to identify the baseline scenario and demonstrate additionality Version 07.0¹¹”. The project activity is a greenfield project. According to the applied methodology for greenfield projects, the baseline scenario is selected from the complete set of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories (Volume 4, Chapter 10, Table 10.17)¹².

¹¹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-02-v7.0.pdf>

¹² https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_10_Ch10_Livestock.pdf

No.	IPCC Alternatives	Applicability	Justification
1.	The manure is collected from the pasture/Range/Paddock.	Not Applicable	Considered not to be realistic alternative, the farms around are all confined farms.
2.	Manure removed from confinement and applied to pasture within 24 hours of excretion	Not Applicable	Not applicable because all the cows and chickens are raised on well-managed animal farms. The manure will be collected from animal farms rather than pastures. Pasture lands also have a limited capacity to apply a high amount of manure.
3.	The manure is disposed by solid storage.	Not Applicable	Not realistic because of the manure removal periods. The alternative requires high storage volumes.
4.	Confinement area where manure can be removed periodically	Not Applicable	Not realistic because the storage of manure in unconfined piles or stacks is not allowed. According to the Air Quality Regulation of Turkey, solid manures shall be stored in impermeable platforms, and measures should be taken to eliminate odour, flies, etc. (article 1.2.4-1.2.6).
5.	The manure is disposed as liquid/slurry.	Not Applicable	Not realistic because the alternative would require huge areas. According to the Air Quality Regulation of Turkey, liquid wastes shall be stored in impervious storage areas and transferred with closed containers.
6.	The manure is disposed in an uncovered anaerobic lagoon;	Applicable	considered as a credible alternative

No.	IPCC Alternatives	Applicability	Justification
7.	The manure is disposed by pit storage below animal confinements.	Not Applicable	Not realistic because the animal farms that supply manure to the project activity do not have the slatted floor through which the manure will be collected.
8.	The manure is disposed by anaerobic digester.	Not Applicable	Not realistic since there is no regulation for the installation of ADs, and also, the AD requires significant investments and know-how.
9.	Excreted dungs on fields are dried and burned as fuel	Not Applicable	Since drying the manure would require a considerable amount of heat to be used as fuel and in many areas in case of sun drying, the alternative is not considered realistic.
10.	The manure is disposed by deep bedding.	Not Applicable	Deep bedding application would require a closed area to prevent any rain and minimise moisture; however, most farms have open spaces. This alternative is not common in Turkey and, therefore, not realistic. Deep bedding is also associated with higher operational costs to keep the bedding clean and non-microbial. This alternative requires more effort and is unrealistic.
11.	Composting via different methods	Not Applicable	Since composting would associate with high investment costs and profound know-how, this alternative is unrealistic. There is also no legal requirement to compost.

No.	IPCC Alternatives	Applicability	Justification
12.	The manure is disposed with litter.	Not Applicable	The farms that supply manure to the Project Activity consist mainly of cattle farms, not poultry farms. As stated in the deep bedding alternative, this alternative would require more effort to implement.
13.	The manure is disposed without litter.	Not Applicable	This alternative would require a proper design and operation associated with effort and more costs for the farmers. This alternative is not realistic.
14.	The manure is disposed via aerobic treatment.	Not Applicable	Since aerobic treatment would associate with high investment costs and high land area requirements, this alternative is unrealistic. There is also no legal requirement for aerobic treatment.
15.	The project activity is undertaken without being registered as a proposed CDM project activity	Not Applicable	Not found attractive as proved by the investment analysis.

Plausible Alternative Scenarios identified are given in the table below:

Alternative Scenario		A realistic Alternative?	Justification
6	The manure is disposed in an uncovered anaerobic lagoon;	Yes	This alternative is considered a realistic alternative. The manure is disposed of in an uncovered anaerobic lagoon;

Step 1b: Consistency with mandatory applicable laws and regulations

The alternative scenario(s) shall be followed with all mandatory applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions

ACM0022 Alternative waste treatment processes Version (03.0)

Alternative Scenario		Compliance with Regulations	Assessment of Compliance
M1	The project activity without being registered as a CDM project activity (i.e. any (combination) of the waste treatment processes listed in Table 2 of ACM0022;	Yes	This scenario is assessed as realistic because there are no mandatory laws or regulations to enforce project registration in the CDM mechanism.
M3	Disposal of the fresh waste in an SWDS without an LFG capture system;	Yes	Since Turkey has no regulation to enforce landfill gas capture, this scenario complies with the current applicable legislation.
P6	Electricity generation in existing and/or new grid-connected electricity plants.	Yes	Since all operating power plants in Turkey shall be connected to the national grid with the approval of the Ministry of Energy, this scenario is also in compliance with all regulations in Turkey.
H4	Use of (an) existing or construction of (a) new on-site or off-site fossil fuel based boiler(s) or air heater(s)	Yes	Since the use of a fossil-based boiler is allowed by the legislation in Turkey, there isn't any non-compliance with this alternative scenario.

ACM0010 GHG emission reductions from manure management systems Version 08.0

Alternative Scenario		Compliance with Regulations	Assessment of Compliance
	The manure is disposed in an uncovered anaerobic lagoon;	Yes	There are no regulations in Turkey that prohibit the disposal of animal manure in an uncovered anaerobic lagoon. So this scenario complies with all laws in Turkey.

All the credible alternatives identified above comply with Turkey's current laws and regulatory requirements. The project activity is not the only alternative scenario that follows all mandatory regulations with which there is general compliance.

Step 2: Barrier Analysis

The investment analysis will probably fully demonstrate and explain the additionality of the project, so step 2 will be skipped

Step 3 – Investment analysis

Step 4 – Common practice analysis

Step 3 and Step 4 are given under Section 3.5 below.

3.5 Additionality

Demonstrate and assess the additionality of the Project, in accordance with the applied methodology and any relevant tools, taking into account the following:

Investment analysis is carried out in line with the methodological tool: “Investment analysis”, Version 11.0¹³.

Three options can be applied for the investment analysis: the simple cost analysis, the investment comparison analysis and the benchmark analysis.

- Option I: Simple cost analysis
- Option II: Investment comparison analysis
- Option III: Benchmark analysis

The simple cost analysis is not applicable for the proposed project because the project activity will have revenue (from electricity sales and waste resources revenues) other than CDM-related income. The investment comparison analysis is also not applicable for the

¹³ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-27-v11.0.pdf>

proposed project because the baseline scenario, providing the same annual electricity output by the Turkish National Grid, is not an investment project.

To conclude, the benchmark analysis will be used to identify whether the proposed project's financial indicators (project IRR in this case) is better than the relevant benchmark value.

With the help of the investment analysis, it shall be demonstrated that the proposed project activity is not economically or financially feasible without the revenue from the sale of VERs. Therefore, the benchmark analysis shall be applied, as there is no alternative project activity for a comparison of the attractiveness of an investment.

Benchmark Analysis

While applying the Benchmark Analysis, Option III, the Project IRR is selected as the financial indicator to demonstrate the project's additionality as stated in the additionality tool.

The Benchmark rate is calculated in line with "Tool 27: Investment Analysis version 11.0", which suggests the applied benchmark shall be appropriate to the type of IRR calculated. Local commercial lending rates, or WACC, are appropriate benchmarks for a project IRR. Required/expected returns on equity are appropriate benchmarks for a project IRR.

The Directorate of Strategy and Budget publishes "Main Economic Indicators" on a monthly basis. Since the project is a mid-term investment (exceeding one year), the medium-term investment lending rate has been selected as the benchmark. As estimated by the Turkish Development Bank, the lending rate for medium-term investment is 14.5 % for December 2019 (the latest available percentage)¹⁴. Thus, 14.5% is taken as the benchmark value for Project IRR (before tax to be conservative).

Assessment of likelihood conditions for each parameter to reach benchmark IRR is provided below:

Electricity Price

To reach the 14.5 % benchmark IRR, electricity prices should increase more than 10 % from the assumed price. Even if the electricity price increases by 10%, project IRR reaches 11.48%, still lower than the benchmark value of 14.5%.

Investment Cost

In order to reach the benchmark IRR, investment costs shall be decreased by more than 10%. Since the equipment contract, which has a higher share of the total costs, is fixed, a 10% decrease in the investment cost is unlikely. Thus, it is not likely for project activity to have a threshold investment cost and reach to benchmark IRR. Even if investment cost decreases by 10%, project IRR reaches 11.65%, still lower than the benchmark value of 14.5%.

¹⁴ https://www.sbb.gov.tr/wp-content/uploads/2020/07/13-faiz_orani-1.xls

Energy Yield

The annual energy yield amount shall increase more than 10 % more than the base case electricity generation amount used in financial analysis to reach the benchmark IRR. Although most biogas power projects use electricity generation amounts from energy yield reports, to be conservative in the financial investment analysis of the project activity, annual electricity generation to be sold, excluding transmission losses, is employed for the analysis. Even if energy yield increases by 10%, project IRR reaches 11.48%, still lower than the benchmark value of 14.5%.

Operation Cost

Operation costs do not significantly impact the project IRR compared to other parameters. Even if annual operation decreases more than 10%, Project IRR reaches 13.31%, still lower than the benchmark IRR.

Item	Value	Units	Source
Installed Power	7.8	MWe	Provisional Acceptance Protocols
The operational lifetime of the project	25	years	Tool to determine the remaining lifetime of the equipment
Net Generation to be sold	32,307,589*(1-1.78%) = 31,731,718 Where 1.78% accounts for the transmission losses	kWh/yr	Capacity Report of the Project
Electricity tariff (for ten years)	133.0	USD/MWh	Please see the IRR calculation sheet
Main Investment Costs	26,642,247.18	EUR	Please see the IRR calculation sheet
Operational Expenses	3,330,306.60	EUR /yr	Please see the IRR calculation sheet.

Calculation and comparison of financial indicators

The technical lifetime of the Ödemis ISWMP Project is determined by using the Tool to determine the remaining lifetime of equipment (v.1). In the tool, it is said that the default lifetime for Diesel/oil/gas fired generator sets is 50,000 hrs (6.89 yrs with 22 hrs per day and

330 days per year). A conservative value of 25 yrs lifetime is considered, and IRR analysis is performed for this period.

For depreciation, the useful year has been taken as 10 years. In addition, a feed-in tariff grants further incentives for five years if local content is sourced from Turkey. Fair value is taken as 5% of the infrastructure cost and is added to the last year of the cash flow in the IRR analysis.

Sensitivity Analysis

The project IRR (after tax) of Ödemis ISWMP Project is calculated based on expected cash flows (investment, operating costs and revenues from waste resources and electricity sales), as used in the financial analysis for the additionality assessment of the project. The parameters and values used for the IRR calculation are available to VVB during validation. The resulting IRR for 25 years is given in the below table.

Table: IRR Results Concerning Different Financial Scenarios

Parameter	Power Price			Investment Cost			Energy Yield			Operating Cost		
Variance	-10%	0%	10%	-10%	0%	10%	-10%	0%	10%	-10%	0%	10%
Project IRR after Tax (for 25 years)	8.68%	10.09%	11.48%	11.65%	10.09%	8.77%	8.68%	10.09%	11.48%	13.31%	10.09%	5.98%

Step 4: Common Practice Analysis

The section below provides the analysis as per step 4 of the “Tool for the demonstration and assessment of additionality”, version 7.0.0 and according to “Common Practice” Tool version 03.¹⁵.

Step 4.1.: Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity: The proposed project has a capacity of 7.8 MWe. Per the guideline of +/-50%, the applicable output range for the project is 3.9 MWe to 11.7 MWe.

Step 4.2.: Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

The projects are located in the applicable geographical area; (Turkey)

¹⁵ Please see [the related link](#).

The projects apply the same measure as the proposed project activity; (alternative waste treatment plant with anaerobic digestion and RDF production)

The projects use the same energy source/fuel and feedstock as the proposed project activity, if the proposed project activity implements a technology switch measure; (Using animal manure and municipal solid wastes as feedstocks)

The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant; (Electricity and heat generation)

The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1; (3.9 MWe to 11.7 MWe)

The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of the proposed project activity, whichever is earlier for the proposed project activity;

Energy Market Regulatory Authority (EPDK) in Turkey publishes and updates the Final List of Renewable Energy Plants Operational in Turkey every year. And with this official list, it is possible to see all renewable energy plants operational till that year, separated by their License Number, Project Owner, Plant Name, Plant Type, Fuel Type, Installed Power, Number of Units Operational and the Start Date of Electricity Generation. This Final List of Renewable Energy Plants Operational is used to determine the Nall number. First, the projects that apply the same measure as the proposed activity are determined from this list (anaerobic digestion with biomass). And then, since it is possible to separate plants according to the fuel type they use, the projects that use the same energy source/fuel and feedstock as the proposed project activity are determined (Using animal manure, municipal solid waste and agriculture residues as feedstocks). It is also possible to separate plants according to the installed capacity; the capacity of the projects within the applicable capacity range of 3.9 MWe to 11.7 MWe is determined. Since it is possible to separate plants according to the start date of operation, the projects that started commercial operation before the start date of the project activity are determined. And finally, from the final determined list, the projects that are neither registered CDM/Gold Standard projects, project activities submitted for registration, nor project activities undergoing validation are determined to note Nall number.

Step 4.3.: within the projects identified in Step 4.2., identify those that are neither registered CDM/Gold Standard projects, project activities submitted for registration, nor project activities undergoing validation. Note their number, Nall.

According to the official report of “Final List on Renewable Energy Plants, 2022¹⁶” published by the Republic of Turkey Energy Market Regulatory Authority (EPDK), the applicable biogas plants (which started operation before the project start date of Ödemis ISWMP Project, which is 29.12.2020) whose anaerobic digesters’ main feedstocks are either animal manure or municipal solid waste or agricultural residues are shown in the table below:

Within the projects identified in the Table above, whose installed capacities are within the proposed project activity’s applicable capacity range calculated in Step 4.1. (3.9 MWe – 11.7 MWe) in 2020, only two of them are neither registered Gold Standard projects, project activities submitted for registration, nor ongoing project validation. The two projects are shown in the below Table.

Accordingly, it is noted that

Nall=2

Step 4.4.: within similar projects identified in Step 4.3., identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number Ndiff.:

Among two projects identified in Step 4.3. all of them are different from Ödemis ISWMP Project since Ödemis ISWMP is an integrated solid waste management facility producing RDF also.

Accordingly, it is noted that

Ndiff=2

Step 4.5: Calculate factor $F=1-Ndiff/Nall$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

In Step 4.3., Nall is noted as 2.

In Step 4.4., Ndiff is noted as 2.

Accordingly, factor $F=1-Ndiff/Nall$ is calculated as follows:

$$F= 1-2/2$$

$$F= 0$$

The paragraph 18 of the methodological tool: “Common practice”, Version 03.0¹⁷ states that

¹⁶ <https://www.epdk.gov.tr/Detay/DownloadDocument?id=YGmoCBgNWRQ=>

¹⁷ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-24-v1.pdf>

“The proposed project activity is a ‘common practice’ within a sector in the applicable geographical area if the factor F is greater than 0.2 and Nall-Ndiff is greater than 3” (pg. 5).

For the proposed project activity, $F=0$ is not greater than 0.2, and also $Nall-Ndiff=0$ which is not greater than 3. Hence, the proposed project activity shall be considered as not a common practice in the region.

Hence, in accordance with the methodological tool: “Combined tool to identify the baseline scenario and demonstrate additionality”, Version 07.0¹⁸, the proposed project activity shall be considered additional.

3.6 Methodology Deviations

There are no methodology deviations applied during this monitoring period.

¹⁸ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-02-v7.0.pdf>

4 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

According to the methodology ACM0022 (version 03.0), baseline emissions are determined as follows:

$$BE_y = \sum_t (BE_{CH_4,t,y} + BE_{WW,t,y} + BE_{EN,t,y} + BE_{NG,t,y}) \times (1 - RATE_{compliance,t})$$

Where:

BE_y	=	Baseline emissions in year y (t CO ₂ e)
$BE_{CH_4,t,y}$	=	Baseline emissions of methane from the SWDS in year y (t CO ₂ e)
$BE_{WW,t,y}$	=	Baseline methane emissions from anaerobic treatment of the wastewater in open anaerobic lagoons or of sludge in sludge pits in the absence of the project activity in year y (t CO ₂ e)
$BE_{EN,t,y}$	=	Baseline emissions associated with energy generation in year y (t CO ₂)
$BE_{NG,t,y}$	=	Baseline emissions associated with natural gas use in year y (t CO ₂)
$RATE_{compliance,t,y}$	=	Discount factor to account for the rate of compliance of a regulatory requirement that mandates the use of alternative waste treatment process t
t	=	Type of alternative waste treatment process

For this Project Activity within the scope of ACM0022 only Baseline methane emissions from the SWDS and Baseline emissions associated with energy generation will be accounted for. The project uses methane from fresh waste produced by anaerobic digestion treatment for electricity&heat generation. It doesn't involve methane emissions from anaerobic wastewater treatment in open anaerobic lagoons or sludge in sludge pits and emissions associated with natural gas use; thus, $BE_{WW,t,y}=0$, $BE_{NG,t,y}=0$. There are also no regulations

that mandate the use of the project activity treatment option in Turkey, So $RATE_{compliance,t,y} = 0$

$$BE_y = BE_{CH_4,y} + BE_{EN,y}$$

Baseline methane emissions from the SWDS ($BE_{CH_4,t,y}$)

Baseline methane emissions from the SWDS are determined using the CDM methodological tool “Emissions from solid waste disposal sites” (version 08.0). The following requirements shall be complied with when applying the tool:

Requirement		Compliance
(a)	$W_{j,x}$ in the tool is the amount of fresh organic waste prevented from disposal in the baseline SWDS due to its treatment in any (combination) alternative waste treatment process	In the Project Activity, anaerobic digestion prevents biodegradable waste from being disposed of in the baseline SWDS. Additionally, in the Material Recovery Facility, plastics, combustibles, paper and cartons are recovered to produce RDF.
(b)	Emission amounts are calculated using Application B in the tool (only fresh waste avoided from disposal after the start of the first crediting period shall be considered)	Application B is applied, and only fresh waste avoided from disposal after the start of the first crediting period is considered.
(c)	Sampling to determine the fractions of different waste types is necessary (note that for the case that the waste is combusted in the project activity, then the parameter $Q_{j,c,y}$ in this methodology is equivalent to the variable $W_{j,x}$ in the tool)	The Project Proponent conducts waste characterisation studies to determine the fractions of different waste types
(d)	The tool instructs that f_y shall be determined based on historical data or contract or regulation requirements specifying the amount of methane that must be destroyed/used (if available). The following additional instruction applies:	The regulation in Turkey does not specify the amount or percentage of LFG that should be destroyed but suggests the installation of a capture system; without

	i. If the regulation requirements specify a percentage of the LFG that is required to be flared, the amount shall equal f_y; ii. If the regulation requirements do not specify the amount or percentage of LFG that should be destroyed but require the installation of a capture system, without requiring the captured LFG to be flared then $f_y = 0$; and iii. If the requirement does not specify any amount or percentage of LFG that should be destroyed but require the installation of a system to capture and flare the LFG, then it is assumed $f_y = 0.2$.	requiring the captured LFG to be flared, f_y is taken as 0.
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$BE_{CH_4,SWDS,y}$ is calculated as per “Tool 04 Emissions from solid waste disposal sites Version 08.0.”

According to Tool 04, the amount of methane in the LFG generated from waste disposal at the SWDS is calculated based on a first-order decay (FOD) model. The model differentiates between the different waste types j with constant decay rates (k_j) and fractions of degradable organic carbon (DOC_j).

$$\left. \begin{array}{l} BE_{CH_4,SWDS,y} \\ PE_{CH_4,SWDS,y} \\ LE_{CH_4,SWDS,y} \end{array} \right\} = \varphi_y \times (1 - f_y) \times GWP_{CH_4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \\
 \times MCF_y \times \sum_{x=1}^y \sum_j (W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times (1 - e^{-k_j}))$$

Where:

$BE_{CH_4,SWDS,y}$	Baseline methane emissions occurring in year y generated from waste disposal at a SWDS during a time period ending in year y (t CO ₂ e/yr)
x	= Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period ($x = 1$) to year y ($x = y$)
y	= Year of the crediting period for which methane emissions are calculated (y is a consecutive period of 12 months)
$DOC_{f,y}$	Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)

$W_{j,x}$ =	Amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x (t)
φ_y =	Model correction factor to account for model uncertainties for year y
f_y =	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
GWP_{CH_4}	= Global Warming Potential for methane
OX	= Oxidation factor (A default value of 0.1 is used)
F	= Fraction of methane in the SWDS gas (volume fraction)
MCF_y	= Methane correction factor for year y
DOC_j	= Fraction of degradable organic carbon in the waste type j (weight fraction)
k	= Decay rate for the waste type j (1/yr)
j	= Type of residual waste or types of waste in the MSW

Determining the parameters required to apply the FOD model

The below table given in Tool 04 is used to determine the parameter for Application B

Parameter	Application B
φ_y	Project or leakage emissions: default values Baseline emissions: default values or project specific value estimated yearly
OX	Default value
F	Default value
$DOC_{f,y}$	In the case of MSW: default value
MCF_y	Default value
k_j	Default value
$W_{j,x}$	Calculated based on monitored data
DOC_j	Default value

f_y	Monitored
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Determining the model correction factor (φ_y)

If baseline emissions are being calculated, then there are two options to determine the φ :

Option 1: Use a default value

Option 2: Determine φ_y based on the specific situation of the project activity

Option 1 is chosen utilising the default values given in the methodological Tool 04 Emissions from solid waste disposal sites. For Application B and for humid/wet conditions, a default value of 0.85 is suggested.

Determining the amounts of waste types j disposed in the SWDS ($W_{j,x}$ or $W_{j,i}$)

Where different waste types j are disposed of or prevented from disposal in the SWDS (for example, in the case of MSW), it is necessary to determine the amount of different waste types ($W_{j,x}$ or $W_{j,i}$). Since the yearly FOD model is applied for the Project Activity, the relevant parameter for the calculations is $W_{j,x}$.

Application A is chosen to determine the amounts of waste types disposed of in the SWDS. $W_{j,x}$ is determined based on information from the SWDS owner. A detailed feasibility study was conducted for the Project Activity, where it is possible to see the amount of all different waste types coming to the SWDS and going into the other units operating in the plant.

Determining the fraction of DOC that decomposes in the SWDS ($DOC_{f,y}$)

As per paragraph 29 of Tool 04, "Emissions from the solid waste disposal sites", Application A is employed where $DOC_{f,y} = DOC_{f,default}$. A default of 0.5 is used for the calculations.

Procedure to determine the methane correction factor (MCF_y)

As per Tool 04, default values (based on SWDS type) can be selected for SWDS without a water table above the bottom of the SWDS. Since no water table above the bottom of the SWDS is present for the project activity, a default of 1.0 is used for Ödemis ISWMP (anaerobic managed solid waste disposal site where there is a controlled placement of waste).

Baseline emissions from generation of energy

This procedure is distinguished depending on whether the baseline is the separate generation of electricity and heat or the combined generation of heat and electricity by cogeneration. The project activity achieves the combined generation of heat and electricity by cogeneration.

Cogeneration of electricity and heat

Baseline emissions from electricity and heat cogeneration are calculated by multiplying electricity generation ($EG_{t,y}$) and the quantity of heat supplied ($HG_{PJ,y}$) with the CO2 emission factor of the fuel used by the cogeneration plant, as follows:

$$BE_{EN,y} = \frac{(EG_{t,y} \times 3.6) \times 10^{-3} + HG_{PJ,y}}{\eta_{cogen}} \times EF_{CO2,BL,CG}$$

Where:

- $BE_{EN,y}$ = Baseline emissions associated with energy generation in year y (t CO2)
- $EF_{CO2,BL,CG}$ = CO2 emission factor of the fossil fuel type used for energy generation by the cogeneration plant in the baseline (t CO2/TJ)
- $HG_{PJ,y}$ = Quantity of heat supplied by the project activity displacing baseline heat generation by a fossil fuel cogeneration plant in year y (TJ)
- $EG_{t,y}$ = Electricity generated by the alternative waste treatment process t and exported to the grid or displacing fossil fuel fired power-only and/or cogeneration captive energy generation in year y (MWh)
- η_{cogen} = Efficiency of the cogeneration plant that would have been used in the absence of the project activity (ratio)

Baseline Emissions due to Animal Manure Use

Baseline Emissions from the use of animal manure in the digesters are calculated as per ACM0010 GHG emission reductions from manure management systems Version 8.0. As per paragraph 26 of ACM0010 Baseline emissions are:

$$BE_y = BE_{CH4,y} + BE_{N2O,y} + BE_{elec/heat,y}$$

Where:

- BE_y = Baseline emissions in year y (t CO2/yr)
- $BE_{CH4,y}$ = Baseline CH4 emissions in year y (t CO2/yr)
- $BE_{N2O,y}$ = Baseline N2O emissions in year y (t CO2/yr)
- $BE_{elec/heat,y}$ = Baseline CO2 emissions from electricity and/or heat used in the baseline (t CO2/yr)

Baseline CH₄ emissions (BE_{CH₄,y})

$$BE_{CH_4,y} = GWP_{CH_4} \times D_{CH_4} \times \sum_{j,LT} (MCF_j \times B_{0,LT} \times N_{LT} \times VS_{LT,y} \times MS\%_{BL,j})$$

Where:

- BE_{CH₄,y} = Baseline CH₄ emissions (t CO₂/yr)
- GWP_{CH₄} = Global Warming Potential (GWP) of CH₄ (t CO₂e/t CH₄)
- D_{CH₄} = Density of CH₄ (t/m³)
- MCF_j = Annual methane conversion factor (MCF) for the baseline AWMS_j
- B_{0,LT} = Maximum methane producing potential of the volatile solid generated by animal type LT (m³CH₄/kg -dm)
- N_{LT} = Annual average number of animals of type LT for the year y (number)
- VS_{LT,y} = Annual volatile solid excretions for livestock LT entering all AWMS on a dry matter weight basis (kg -dm/animal/yr)
- MS%_{BL,j} = Fraction of manure handled in system j in the baseline
- LT = Type of livestock
- j = Type of treatment system

To determine VS_{LT,y} paragraph 30, Option 3 of ACM0010 is applied: Scaling default IPCC values VS_{default} to adjust for a site-specific average animal weight as shown in equation below.

$$VS_{LT,y} = \left(\frac{W_{site}}{W_{default}} \right) \times VS_{default} \times nd_y$$

Where:

- VS_{LT,y} = Annual volatile solid excretions for livestock LT entering all AWMS on a dry matter weight basis (kg -dm/animal/yr)
- W_{site} = Average animal weight of a defined livestock population at the project site (kg)
- W_{default} = Default average animal weight of a defined population (kg)

VS_{default} = Default value for the volatile solid excretion per day on a dry matter basis for a defined livestock population (kg-dm/animal/day)

nd_y = Number of days treatment plant was operational in year y

Annual average number of animals of type LT (N_{LT}) is determined as per Option 1 of ACM0010 employing;

$$N_{LT} = N_{da,LT} \times \left(\frac{N_{p,LT}}{365} \right)$$

Where:

N_{LT} = Annual average number of animals of type LT for the year y (number)

$N_{da,LT}$ = Daily stock of animals of type LT in the farm, discounting dead and discarded animals (number)

$N_{p,LT}$ = Number of animals of type LT produced annually for the year y (number)

Baseline N2O emissions ($BE_{N2O,y}$)

$$BE_{N2O,y} = GWP_{N2O} \times CF_{N2O-N,N} \times \frac{1}{1000} \times (E_{N2O,D,y} + E_{N2O,ID,y})$$

Where:

$BE_{N2O,y}$ = Annual baseline N2O emissions in (t CO2e/yr)

GWP_{N2O} = Global Warming Potential (GWP) for N2O (t CO2e/tN2O)

$CF_{N2O-N,N}$ = Conversion factor N2O-N to N2O (44/28)

$E_{N2O,D,y}$ = Direct N2O emission in year y (kg N2O-N/year)

$E_{N2O,ID,y}$ = Indirect N2O emission in year y (kg N2O-N/year)

$$E_{N2O,D,y} = \sum_{j,LT} EF_{N2O,D,j} \times NEX_{LT,y} \times N_{LT} \times MS\%_{Bl,j}$$

Where:

$E_{N2O,D,y}$ = Direct N2O emission in year y (kg N2O-N/yr)

$EF_{N2O,D,j}$ = Direct N2O emission factor for the treatment system j of the manure management system (kg N2O-N/kg N)

$NEX_{LT,y}$ = Annual average nitrogen excretion per head of a defined livestock population (kg N/animal/yr) estimated as described in appendix 2

N_{LT} = Annual Average number of animals of type LT for the year y

$MS\%_{BI,j}$ = Fraction of manure handled in system j (fraction)

$$E_{N2O,ID,y} = \sum_{j,LT} EF_{N2O,ID} \times F_{gasMS,j,LT} \times NEX_{LT,y} \times N_{LT} \times MS\%_{BI,j}$$

Where:

$E_{N2O,ID,y}$ = Indirect N2O emission in year y (kg N2O-N/year)

$EF_{N2O,ID}$ = Indirect N2O emission factor for N2O emissions from atmospheric deposition of nitrogen on soils and water surfaces (kgN2O-N/kg NH3-N and NOX-N)

$F_{gasMS,j,LT}$ = Default values for nitrogen loss due to volatilisation of NH3 and NOX from manure management (fraction)

$NEX_{LT,y}$ = Annual average nitrogen excretion per head of a defined livestock population (kg N/animal/year) estimated as described in appendix 2

N_{LT} = Annual average number of animals of type LT for the year y

$MS\%_{BI,j}$ = Fraction of manure handled in system j (fraction)

4.2 Project Emissions

The project emissions in year y are calculated for each alternative waste treatment process implemented in the project activity as follows:

$$PE_y = PE_{COMP,y} + PE_{AD,y} + PE_{GAS,y} + PE_{RDF_SB,y} + PE_{INC,y}$$

Where:

PE_y = Project emissions in year y (t CO2e)

$PE_{COMP,y}$ = Project emissions from composting or co-composting in year y (t CO2e)

$PE_{AD,y}$ = Project emissions from anaerobic digestion and biogas combustion in year y (t CO₂e)

$PE_{GAS,y}$ = Project emissions from gasification in year y (t CO₂e)

$PE_{RDF_SB,y}$ = Project emissions associated with RDF/SB in year y (t CO₂e)

$PE_{INC,y}$ = Project emissions from incineration in year y (t CO₂e)

Project emissions from composting or co-composting are not accounted for since there is no composting activity in Ödemiş ISWMP.

Project emissions from gasification are not accounted for since no gasification activity exists in Ödemiş ISWMP.

Project emissions from incineration are not accounted for since there is no incineration activity in Ödemiş ISWMP.

Project emissions from anaerobic digestion ($PE_{AD,y}$)

$PE_{AD,y}$ is calculated according to “TOOL14: Project and leakage emissions from anaerobic digesters”.

As per paragraph 13 of the “TOOL14: Project and leakage emissions from anaerobic digesters”, The project emissions associated with the anaerobic digester ($PE_{AD,y}$) are determined as follows:

$$PE_{AD,y} = PE_{EC,y} + PE_{FC,y} + PE_{CH_4,y} + PE_{flare,y}$$

Where:

$PE_{AD,y}$ = Project emissions associated with the anaerobic digester in year y (t CO₂e)

$PE_{EC,y}$ = Project emissions from electricity consumption associated with the anaerobic digester in year y (t CO₂e)

$PE_{FC,y}$ = Project emissions from fossil fuel consumption associated with the anaerobic digester in year y (t CO₂e)

$PE_{CH_4,y}$ = Project emissions of methane from the anaerobic digester in year y (t CO₂e)

$PE_{flare,y}$ = Project emissions from flaring of biogas in year y (t CO₂e)

Step 1: Determination of the quantity of methane produced in the digester ($Q_{CH_4,y}$)

There are two different procedures to determine the quantity of methane produced in the digester in year y ($Q_{CH_4,y}$). For large-scale projects, only Option 1 is used.

Option 1: Procedure using monitored data $Q_{CH_4,y}$ shall be measured using the “Tool 08 Tool to determine the mass flow of a greenhouse gas in a gaseous stream” (version 03.0). When applying the tool, the following applies:

- (a) The gaseous stream to which the tool is applied is the biogas collected from the digester;
- (b) CH_4 is the greenhouse gas i for which the mass flow should be determined; and
- (c) The flow of the gaseous stream should be measured hourly or at a smaller time interval and then accumulated for the year y . Please note that units need to be converted to tons when applying the results in this tool.

Option C is chosen for determining the mass flow of CH_4 in biogas resulting from the anaerobic digesters.

As per paragraph 29 of the Tool to determine the mass flow of greenhouse gas in a gaseous stream, the mass flow of CH_4 ($F_{CH_4,t}$) is determined as follows:

$$F_{i,t} = V_{t,wb,n} \times v_{i,t,wb} \times \rho_{i,n}$$

With:

$$\rho_{i,n} = \frac{P_n \times MM_i}{R_u \times T_n}$$

Where:

- $F_{i,t}$ = Mass flow of greenhouse gas i in the gaseous stream in time interval t (kg gas/h)
- $V_{t,wb,n}$ = Volumetric flow of the gaseous stream in time interval t on a wet basis at normal conditions (m^3 wet gas/h)
- $v_{i,t,wb}$ = Volumetric fraction of greenhouse gas i in the gaseous stream in time interval t on a wet basis (m^3 gas i/m^3 wet gas)
- $\rho_{i,n}$ = Density of greenhouse gas i in the gaseous stream at normal conditions (kg gas i/m^3 wet gas i)
- P_n = Absolute pressure at normal conditions (Pa)
- T_n = Temperature at normal conditions (K)
- MM_i = Molecular mass of greenhouse gas i (kg/kmol)
- R_u = Universal ideal gases constant (Pa.m³ /kmol.K)

As per paragraph 30 of the Tool to determine the mass flow of a greenhouse gas in a gaseous stream, the following equation should be used to convert the volumetric flow of the gaseous stream from actual conditions to normal conditions of temperature and pressure:

$$V_{t,wb,n} = V_{t,wb} \times [(T_n/T_t) \times (P_t/P_n)]$$

Where:

$V_{t,wb,n}$ = Volumetric flow of the gaseous stream in a time interval t on a wet basis at normal conditions (m^3 wet gas/h)

$V_{t,wb}$ = Volumetric flow of the gaseous stream in time interval t on a wet basis (m^3 wet gas/h)

P_t = Pressure of the gaseous stream in time interval t (Pa)

T_t = Temperature of the gaseous stream in time interval t (K)

P_n = Absolute pressure at normal conditions (Pa)

T_n = Temperature at normal conditions (K)

Step 2: Determination of project emissions from electricity consumption ($PE_{EC,y}$)

Option 1 is chosen to determine project emissions from electricity consumption. As per Option 1 of Tool 14, $PE_{EC,y}$ shall be calculated using the “Tool 05 Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” version 03.0, where the project emission source j referred to in the tool is the total electricity consumption associated with the anaerobic digestion facility.

According to Paragraph 16 of Tool 05, project emissions from the consumption of electricity are calculated based on the quantity of electricity consumed, an emission factor for electricity generation and a factor to account for transmission losses, as follows:

$$PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EF,j,y} \times (1 + TDL_{j,y})$$

Where:

$PE_{EC,y}$ = Project emissions from electricity consumption in year y (t CO₂ / yr)

$EC_{PJ,j,y}$ = Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)

$EF_{EF,j,y}$ = Emission factor for electricity generation for source j in year y (t CO₂/MWh)

$TDL_{j,y}$ = Average technical transmission and distribution losses for providing electricity to source j in year y

j = Sources of electricity consumption in the project

Determination of the emission factor for electricity generation $EF_{EF,j,y}$

according to the tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0), for the project in the case of Scenario A: Electricity consumption from the grid, the project participants choose Option A1: Calculate the combined margin emission factor of the applicable electricity system, using the procedures in the latest approved version of the “Tool to calculate the emission factor for an electricity system” (version 07.0)

For the emission factor, the Turkish Ministry of Energy and Natural Resources publication, which indicates Turkey's National Electric Grid Emission Factor for the year 2020, was used. The publication includes calculated Emission Factor values that are Operating Margin (OM), Growth Based Margin (Build Margin-BM) and Combined Margin (CM) Emission Factors for the relevant year with the usage of the IPCC's Clean Development Methodology Tool 07-v07.0. For this calculation, information regarding the data set is given below in detail;

- TEİAŞ Turkey's electricity generation-consumption and loss statistics,
- Common prepared report under Turkey's National Greenhouse Gas Inventory Reporting Format. - Common Reporting Format (CRF) tables for electricity generation (1.A.1.a.i) emission values
- Chronological order of power generation plants from TEİAŞ Load Dispatch Department with commissioning dates, plant names, fuel types, installed power values, electricity generation for the calculated year
- Checking off Volunteers from the websites of Gold Standard (GS), Verified Carbon Standard (VCS) and Global Carbon Council for the ownership status of the carbon reduction certificate and,
- From Clean Development Mechanism (CDM) Tool 009- V03.0, Power plant efficiency figures are used

According to this publication;

- Operating Margin-OM; 0.7424 tCO₂/MWh
- Build Margin-BM; 0.3680 tCO₂/MWh

- Combined Margin-CM (other renewables than solar and wind); 0.5552 tCO₂/MWh¹⁹

Step 3: Determination of project emissions from fossil fuel consumption (PE_{FC,y})

The anaerobic digester facility of the project does not use fossil fuels, thus,

$$PE_{FC,y} = 0$$

Step 4: Determination of project emissions of methane from the anaerobic digester (PE_{CH₄,y})

As per paragraph 23 of Tool 14, Project Emissions of Methane from the Anaerobic Digester include emissions during maintenance of the digester, physical leaks through the roof and side walls, and release through safety valves due to excess pressure in the digester. These emissions are calculated using a default emission factor ($EF_{CH_4,ult}$), as follows:

$$PE_{CH_4,y} = Q_{CH_4,y} \times EF_{CH_4,default} \times GWP_{CH_4}$$

Where:

PE _{CH₄,y}	=	Project emissions of methane from the anaerobic digester in year y (t CO ₂ e)
Q _{CH₄,y}	=	Quantity of methane produced in the anaerobic digester in year y (t CH ₄)
EF _{CH₄,default}	=	Default emission factor for the fraction of CH ₄ produced that leaks from the anaerobic digester (fraction)
GWP _{CH₄}	=	Global warming potential of CH ₄ (t CO ₂ / t CH ₄)

Step 5: Determination of project emissions from flaring of biogas (PE_{flare,y})

As per paragraph 24 of Tool 14, If the project activity includes flaring of biogas, then project emissions from flaring of biogas (PE_{flare,y}) shall be estimated using the “Tool 06 project emissions from flaring” version 04.0. The following applies:

- For small scale projects, project participants may adopt a default value for the fraction of methane in the biogas ($f_{CH_4,ult}$) in applying the tool; and
- The tool provides default factors for the flare efficiency, which can be used for large or small scale projects as described in the tool.

According to “Tool 06 project emissions from flaring” version 04.0, the project emissions calculation procedure is given in the following steps:

- (a) STEP 1: Determination of the methane mass flow of the residual gas;
- (b) STEP 2: Determination of the flare efficiency;
- (c) STEP 3: Calculation of project emissions from flaring.

The calculation procedure in this tool determines the methane flow rate before and after the destruction in the flare, taking into account the amount of air supplied to the combustion reaction and the exhaust gas composition (oxygen and methane).

The calculation procedure in this tool determines the project emissions from flaring the residual gas ($PE_{\text{flare},y}$) based on the flare efficiency ($\eta_{\text{flare},m}$) and the mass flow of methane to the flare ($F_{\text{CH}_4,\text{RG},m}$). The flare efficiency is determined for each minute m of year y based on monitored data or default values.

The project activity applies an enclosed flare. The temperature in the exhaust gas of the flare is measured to determine whether the flare is operating or not.

STEP 1: Determination of the methane mass flow of the residual gas

TOOL08 shall be used to determine the mass flow of methane in the residual gaseous stream in the minute m

The following requirements apply:

- (a) Tool 08 shall be applied to the residual gas;
- (b) The flow of the gaseous stream shall be measured continuously;
- (c) CH₄ is the greenhouse gas i for which the mass flow should be determined;
- (d) The simplification offered for calculating the molecular mass of the gaseous stream is valid (equations 3 and 17 in the Tool 08); and
- (e) The time interval t for which mass flow should be averaged is every minute m .

The calculation follows the procedure as described by the “Tool to determine the mass flow of a greenhouse gas in a gaseous stream”. Option C is applied: Same basis (wet basis) is considered for the measurement of the volumetric flow rate of the residual gas and the measurement of the volumetric fraction of methane in the residual gas.

The mass flow of greenhouse gas i ($F_{\text{CH}_4,t}$) is determined as follows:

$$F_{i,t} = V_{t,wb,n} \times v_{i,t,wb} \times \rho_{i,n}$$

With

$$\rho_{i,n} = \frac{P_n \times MM_i}{R_u \times T_n}$$

Where

$F_{CH_4,t}$	Mass flow of greenhouse gas (CH ₄) in the gaseous stream in time interval t (kg gas/h)
$V_{t,wb}$	Volumetric flow of the gaseous stream in time interval t on a wet basis (m ³ wet gas/h)
$V_{CH_4,wb,t}$	Volumetric fraction of greenhouse gas CH ₄ in the gaseous stream in a time interval t on a wet basis (m ³ gas i / m ³ wet gas)
$\rho_{CH_4,n}$	Density of greenhouse gas CH ₄ in the gaseous stream at normal conditions t (kg gas i/m ³ wet gas i)
P_n	Absolute pressure of the gaseous stream at normal conditions (Pa)
MM_i	Molecular mass of greenhouse gas i (kg/kmol)
R_u	Universal ideal gases constant (Pa.m ³ /kmol.K)
T_n	Temperature of the gaseous stream at normal conditions (K)

STEP 2. Determination of flare efficiency

The determination of the minutely flare efficiency depends on the operation of the flare (e.g. temperature), the type of flare used (open or enclosed) and, in the case of enclosed flares, the approach selected by project participants to determine the flare efficiency (default value or continuous monitoring).

In the case of the Ödemis ISWMP Project, a semi-enclosed flare is used, and the flare efficiency is determined by default value, thus Option A. For enclosed flares that are defined as low height flares, the flare efficiency in the minute m ($\eta_{\text{flare},m}$) shall be adjusted, as a conservative approach, by subtracting 0.1 from the efficiency as determined in Options A or B. The project activity employs a semi-enclosed flare unit with a total height of 7m and a diameter of 1.27m; so it is a high-height flare²⁰. The flare unit operates with a maximum flow rate of 1500 m³/hr. Moreover, the combustion temperature of the flare unit is in the range of 850°C – 1200°C.



Option A: Default value

In the case of enclosed flares and the use of the default value for the flare efficiency, the flare efficiency in the minute m ($\eta_{\text{flare},m}$) is 90% when the following two conditions are met to demonstrate that the flare is operating:

- The temperature of the flare ($T_{\text{EG},m}$) and the flow rate of the residual gas to the flare ($F_{\text{RG},m}$) is within the manufacturer's specifications for the flare ($\text{SPEC}_{\text{flare}}$) in minute m ; and
- The flame is detected in minute m (Flame_m)

²⁰ Flare Specifications are provided to the VVB for review. Please also see: <https://www.progecosrl.com/en/emr-advanced-biogas-combustion-flare.html>

Otherwise, ($\eta_{\text{flare},m}$) is 0 %. For this Project Activity, to be conservative, a flare efficiency of 0% is adopted.

STEP 3. Calculation of project emissions from flaring

Project emissions from flaring are calculated as the sum of emissions from each minute m , based on the methane mass flow in the residual gas ($F_{\text{CH}_4,\text{RG},m}$) and the flare efficiency

($\eta_{\text{flare},m}$), as follows:

$$PE_{\text{flare},y} = GWP_{\text{CH}_4} \times \sum_{m=1}^{525600} F_{\text{CH}_4,\text{RG},m} \times (1 - \eta_{\text{flare},m}) \times 10^{-3}$$

Where:

$PE_{\text{flare},y}$ Project emissions from flaring of the residual gas in year y (tCO₂e)

$F_{\text{CH}_4,\text{RG},m}$ Mass flow of methane in the residual gas in the minute m (kg)

$\eta_{\text{flare},m}$ Flare efficiency in minute m

4.3 Leakage

According to the ACM0022 (Version 03.0), leakage emissions are associated with composting/co-composting, anaerobic digestion and the use of RDF/SB that is exported outside the project boundary. Leakage emissions are determined as follows:

$$LE_y = LE_{\text{COMP},y} + LE_{\text{AD},y} + LE_{\text{RDF_SB},y}$$

Where:

LE_y = Leakage emissions in the year y (t CO₂e)

$LE_{\text{COMP},y}$ = Leakage emissions from composting or co-composting in year y (t CO₂e)

$LE_{\text{AD},y}$ = Leakage emissions from anaerobic digester in year y (t CO₂e)

$LE_{\text{RDF_SB},y}$ = Leakage emissions associated with RDF/SB in year y (t CO₂e)

For this project activity, the waste by-products of the project do not involve compost or co-compost and the use of RDF/SB outside of the project boundary. Leakage emissions associated with anaerobic digestion of waste ($LE_{\text{AD},y}$) are calculated according to “TOOL14: Project and leakage emissions from anaerobic digesters” (Version 02.0). According to the tool, the leakage emissions associated with the anaerobic digester ($LE_{\text{AD},y}$) depend on how the digestate is managed. They include emissions associated with storage and composting

of the digestate. Since the solid digestate will be outward transported and landfilled and will not be stored under anaerobic conditions, and the wastewater will be treated aerobically by a wastewater treatment plant nearby, leakage emission is zero, $LE_{AD,y}=0$.

Leakage emissions associated with RDF/SB ($LE_{RDF_SB,y}$)

As per paragraph 90 of ACM0022, Leakage emissions associated with RDF/SB are accounted for by the organic waste byproducts of the treatment process (not by-products from the RDF/SB combustor), which may be composted or disposed of in an SWDS, and the end-use of RDF/SB that is exported off-site, as follows:

$$LE_{RDF_SB,y} = LE_{ENDUSE_RDF_SB,y} + LE_{SWDS,WBP_RDF_SB,y}$$

Where:

$LE_{RDF_SB,y}$ = Leakage emissions associated with RDF/SB in year y (t CO₂e)

$LE_{SWDS,WBP_RDF_SB,y}$ = Leakage emissions associated with disposing of waste byproducts associated with RDF/SB production in an SWDS in year y (t CO₂e)

$LE_{ENDUSE_RDF_SB,y}$ = Leakage emissions associated with the end-use of RDF/SB exported outside the project boundary in year y (t CO₂e)

In the Project Activity, produced RDF will be sent to the cement factories and not disposed of in an SWDS; for this reason, $LE_{SWDS,WBP_RDF_SB,y}$ is 0.

Leakage emissions associated with end use of RDF/SB exported outside the project boundary ($LE_{ENDUSE,RDF_SB,y}$)

The potential leakage emissions associated with using the RDF/SB exported outside the project boundary result from the combustion activities in cement factories. For this Project Activity option (b) End-use 2 is applicable

Leakage emissions from combusted off-site end use of RDF/SB ($LE_{ENDUSE,RDF_SB,y}$) (End use 2)

as per the procedure explained in Paragraph 94 of the ACM0022. This procedure estimates emissions associated with the combustion of RDF/SB outside the project boundary, where the combustor is outside the control of the project participants. Carbon dioxide emissions ($LE_{ENDUSE,RDF_SB,y}$) are calculated as follows:

$$LE_{ENDUSE_RDF_SB,y} = Q_{RDF_SB,COM,y} \times NCV_{RDF_SB,y} \times EF_{CO2_RDF_SB,y}$$

Where:

$LE_{ENDUSE_RDF_SB,y}$	=	Leakage emissions of CO ₂ from off-site combustion of RDF/SB in year y (t CO ₂)
$Q_{RDF_SB,COM,y}$	=	Quantity of RDF/SB exported off-site with potential to be combusted in year y (t)
$NCV_{RDF_SB,y}$	=	Net calorific value of RDF/SB in year y (GJ/t)
$EF_{CO_2_RDF_SB,y}$	=	CO ₂ emissions factor for RDF/SB in year y (t CO ₂ /GJ)

4.4 Estimated Net GHG Emission Reductions and Removals

To calculate the emission reductions the project participant shall apply the following equation:

$$ER_y = BE_y - PE_y - LE_y$$

Where:

ER_y = Emissions reductions in year y (t CO₂e)

BE_y = Baseline emissions in year y (t CO₂e)

PE_y = Project emissions in the year y (t CO₂e)

LE_y = Leakage emissions in year y (t CO₂e)

Calculation of Baseline Emissions

$$BE_y = BE_{CH_4,y} + BE_{EN,y}$$

For the calculation of $BE_{CH_4,y}$

$$\left. \begin{matrix} BE_{CH_4,SWDS,y} \\ PE_{CH_4,SWDS,y} \\ LE_{CH_4,SWDS,y} \end{matrix} \right\} = \varphi_y \times (1 - f_y) \times GWP_{CH_4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y}$$

$$\times MCF_y \times \sum_{x=1}^y \sum_j (W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times (1 - e^{-k_j}))$$

Used Default Parameters	Applied value
φ_y	85%
OX	0.1
F	50%
$DOC_{f,y}$	50%
MCF_y	1
f_y	0

Using the above default values given in the table

$$\varphi_y \times (1 - f_y) \times GWP_{CH_4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} = 7.14$$

$$\times MCF_y$$

WASTE COMPOSITION & DEGRADABLE ORGANIC CONTENT – DOCj & DECAY RATE kJ						
	Food	Paper & Card board	Wood	Textile	Garden, Yard and Park Waste	Inerts Plastic Metals Others
Waste Characterisation Study conducted for 2019	51.12%	7.32%	0.00%	24.68%	1.68%	13.74%
Degradable Organic Content DOCj	15.0%	40.0%	43.0%	24.0%	20.0%	0.0%
Decay Rate - kj	0.185	0.060	0.030	0.060	0.100	-

BASELINE PER TON OF WASTE DEPOSITED ON SITE			
Per t waste	wj	DOCj	kj
Food	0.51	15.0%	0.185
Paper and Cardboard	0.07	40.0%	0.06

Wood	0.00	43.0%	0.03
Textile	0.25	24.0%	0.06
Garden and Park Waste	0.02	20.0%	0.1
Inorganic and other	0.14	0.0%	0

Year	1	2	3	4	5	6	7
$\Sigma W_{j,x} \cdot \text{DOC}_j \cdot e^{-kj} \cdot (y-x)^{(1-e^{-kj})}$	0.01842 28	0.01590 51	0.01377 72	0.01197 54	0.01044 66	0.00914 66	0.00803 85
$\phi y(1-f_y) \cdot \text{GWPCH}_4 \cdot (1-\text{OX}) \cdot 16/12 \cdot F \cdot \text{DOC}_{f,y} \cdot \text{MCF}_y \cdot \Sigma W_{j,x} \cdot \text{DOC}_j \cdot e^{-kj} \cdot (y-x)^{(1-e^{-kj})}$	0.13153 89	0.11356 26	0.09836 92	0.08550 42	0.07458 86	0.06530 67	0.05739 52
Disposed Waste Amount (tons)	268,434	269,936	271,400	273,015	274,399	276,027	277,422
$\text{BECH}_4, \text{SWDS}_y$	35,309	65,991	92,759	116,238	136,898	155,196	171,453

For the calculation of $\text{BE}_{\text{EN},y}$

$$\text{BE}_{\text{EN},y} = \frac{(EG_{t,y} \times 3.6) \times 10^{-3} + HG_{PJ,y}}{\eta_{\text{cogen}}} \times EF_{\text{CO}_2, \text{BL}, \text{CG}}$$

	Value Applied	Unit	Reference
$\text{BE}_{\text{EN},y}$ =	15,937	tCO ₂	Calculated
$EF_{\text{CO}_2, \text{BL}, \text{CG}}$ =	58.3	tCO ₂ /TJ	IPPC 2006 Guidelines Table 1.4 Natural Gas Upper Value
$HG_{PJ,y}$ =	13660714*5.5*0.421*(3.6*10 ⁻⁶)	TJ	Data is Obtained from Capacity Report

$EG_{t,y}$	=	32,307.6	MWh	Data is Obtained from Capacity Report
η_{cogen}	=	84.2%	-	Manufacturer's specifications

Calculation of Baseline Emissions due to Animal Manure Usage in the Digester

$$BE_{Manure,y} = BE_{CH_4,y} + BE_{N_2O,y}$$

Calculation of Baseline Methane Emissions

$$BE_{CH_4,y} = GWP_{CH_4} \times D_{CH_4} \times \sum_{j,LT} (MCF_j \times B_{0,LT} \times N_{LT} \times VS_{LT,y} \times MS\%_{Bl,j})$$

	Value Applied	Unit	Reference
$BE_{CH_4,y}$	= 23,631.44	tCO2	Calculated
GWP_{CH_4}	= 28	tCO_2e/tCH_4	IPCC – AR5 Climate Change 2013: Chapter 2 – Table 2.14
D_{CH_4}	= 0.00067	t/m^3	ACM0010
MCF_j	= 0.73		2019 IPCC Refinement, Table 10.17, p. 67 (uncovered anaerobic lagoon for 17.9°C, warm temperate moist)
$B_{0,LT}$	= Dairy Cattle = 0.24 Other Cattle = 0.17 Poultry = 0.36	$m^3CH_4/kg - dm$	2019 IPCC Refinement, Table 10.16, p. 10.6
N_{LT}	= Dairy Cattle = 2,345 Other Cattle = 600 Poultry = 3,500	number	Project Proponent
$VS_{LT,y}$	= Dairy Cattle = 10.7 Other Cattle = 14.1 Poultry = 14.2	$kg - dm/animal/yr$	2019 IPCC Refinement, Table 10.13A p. 10.55 Middle East
$MS\%_{Bl,j}$	= 100%		

Calculation of Baseline N₂O Emissions

$$BE_{N_2O,y} = GWP_{N_2O} \times CF_{N_2O-N,N} \times \frac{1}{1000} \times (E_{N_2O,D,y} + E_{N_2O,ID,y})$$

	Value Applied	Unit	Reference
BE_{N₂O,y}	644.88	tCO ₂ e	Calculated
GWP_{N₂O}	265	tCO ₂ e/t N ₂ O	IPCC – AR5 Climate Change 2013: Chapter 2 – Table 2.14
CF_{N₂O-N,N}	1.57	-	44/28 ACM0010
E_{N₂O,D,y}	3.30	kg N ₂ O – N/year	Calculated
E_{N₂O,ID,y}	1545.31	kg N ₂ O – N/year	Calculated

Calculation of Direct N₂O Emissions

$$E_{N_2O,D,y} = \sum_{j,LT} EF_{N_2O,D,j} \times NEX_{LT,y} \times N_{LT} \times MS\%_{Bl,j}$$

		Value Applied		Unit	Reference
$E_{N2O,D,y}$	=	3.30		$kg\ N2O - N/year$	Calculated
$EF_{N2O,D,j}$	=	EF3 Cattle	0.000	$kg\ N2O - N / kg\ NH3 - N$ <i>and</i> $NOx - N$	2019 IPCC Refinement, Table 10.21, p. 10.90. For cattle manure under uncovered anaerobic lagoon the EF3 value is 0.00, for poultry manure with or without litter EF3 value is 0.001
		EF3 Poultry	0.001		
$NEX_{LT,y}$	=	Dairy Cattle	=0.50	$kg\ N / 1000kg\ animal\ mass/yr$	2019 IPCC Refinement, Table 10.19, Middle East, Mean
		Other cattle	=0.55		
		Poultry	=1.29		
N_{LT}	=	Dairy Cattle	=2,345	number	Project Proponent
		Other cattle	=600		
		Poultry	=3,500		
$MS\%_{BI,j}$	=	100%			Project Proponent

Calculation of Indirect N₂O Emissions

$$E_{N_2O,ID,y} = \sum_{j,LT} EF_{N_2O,ID} \times F_{gasMS,j,LT} \times NEX_{LT,y} \times N_{LT} \times MS\%_{BI,j}$$

		Value Applied		Unit	Reference
$E_{N2O,ID,y}$	=	1,545.31		$kg\ N2O - N/year$	
$EF_{N2O,ID}$	=	EF4	0.010	$kg\ N2O - N / kg\ NH3 - N$ $and\ NOx - N$	EF4: Emission factor for N2O emissions from N volatilisation and redeposition
		EF5	0.011		EF5: Emission factor for N2O emissions from N leaching and runoff
$F_{gasMS,j,LT}$	=	0.21		fraction	2019 IPCC Refinement, Table 11.3, p. 11.26
$NEX_{LT,y}$	=	Dairy Cattle	=0.50	$kg\ N / 1000kg\ animal\ mass/yr$	2019 IPCC Refinement, Table 10.19, Middle East, Mean, Page 10.83
		Other cattle	=0.55		
		Poultry	=1.29		
N_{LT}	=	Dairy Cattle	=2,345	number	Project Proponent
		Other cattle	=600		
		Poultry	=3,500		
$MS\%_{BI,j}$	=	100%		fraction	Project Proponent

Calculation of Project Emissions

$$PE_y = PE_{COMP,y} + PE_{AD,y} + PE_{GAS,y} + PE_{RDF,SB,y} + PE_{INC,y}$$

For the Project Activity $PE_y = PE_{AD,y}$

Calculation of Project emissions from anaerobic digestion ($PE_{AD,y}$)

$$PE_{AD,y} = PE_{EC,y} + PE_{FC,y} + PE_{CH_4,y} + PE_{flare,y}$$

For the Project Activity $PE_{AD,y} = PE_{EC,y} + PE_{CH_4,y} + PE_{flare,y}$

$$PE_{CH_4,y} = Q_{CH_4,y} \times EF_{CH_4,default} \times GWP_{CH_4}$$

	Applied Value	Unit	Reference
$PE_{CH_4,y}$ =	3,570.92	tCO ₂	Calculated
$Q_{CH_4,y}$ =	4,554.75	tCH ₄	Calculated
$EF_{CH_4,default}$ =	0.0028	fraction	Tool "Project and leakage emissions from anaerobic digesters",
GWP_{CH_4} =	28	t CO ₂ / t CH ₄	IPCC – AR5 Climate Change 2013: Chapter 2 – Table 2.14

$$F_{i,t} = V_{t,wb,n} * V_{i,wb,t} * \rho_{i,n}$$

where $Q_{CH_4,y} = F_{i,t}$

	Applied Value	Unit	Reference
$F_{i,t} = Q_{CH_4,y}$ =	4,554.75	tCH ₄	Calculated
$V_{t,wb,n}$ =	13,660,714	m ³ wet gas/y	Capacity Report of the Project Activity
$V_{i,wb,t}$ =	0.50	m ³ gas i/m ³ wet gas	ACM0010
ρ_{CH_4} =	0.00067	ton CH ₄ /m ³ CH ₄	ACM0010

Project Emissions due to Electricity Use

$$PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EF,j,y} \times (1 + TDL_{j,y})$$

	Applied Value	Unit	Reference
$PE_{EC,y}$ =	2,168.48	tCO ₂	Calculated
$EC_{PJ,j,y}$ =	3,792	MWh	Capacity Report of the Project Activity
$EF_{EF,j,y}$ =	0.5552	tCO ₂ /MWh	Ministry of Energy
$TDL_{j,y}$ =	3%	-	Methodological tool "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0)

Project Emissions due to flaring of Biogas

$$PE_{flare,y} = GWP_{CH_4} \times \sum_{m=1}^{525600} F_{CH_4,RG,m} \times (1 - \eta_{flare,m}) \times 10^{-3}$$

	Applied Value	Unit	Reference
$PE_{flare,y} =$	750.4	tCO2e	Calculated
$GWP_{CH_4} =$	28	tCO2e/tCH4	IPCC – AR5 Climate Change 2013: Chapter 2 – Table 2.14
$F_{CH_4,RG,m} =$	26.8	tCH4	= 80,000*0.50*0.00067
$\eta_{flare,m} =$	0		Tool "Project emissions from flaring"

Leakage Emissions

$$LE_{ENDUSE_RDF_SB,y} = Q_{RDF_SB,COM,y} \times NCV_{RDF_SB,y} \times EF_{CO_2_RDF_SB,y}$$

	Applied Value	Unit	Reference
$LE_{ENDUSE_RDF_SB,y} =$	5,228	tCO2	Calculated
$Q_{RDF_SB,COM,y} =$	2,400	ton	Project Proponent logbook in 2022 an average of 200 ton/month RDF produced sent to the cement factories
$NCV_{RDF_SB,y} =$	18	GJ/ton	table 1.2 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines
$EF_{CO_2_RDF_SB,y} =$	0.121	tCO2/GJ	table 1.2 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2021	75,522	6,490	5,228	63,804
2022	106,204	6,490	5,228	94,486
2023	132,972	6,490	5,228	121,254
2024	156,451	6,490	5,228	144,733
2025	177,111	6,490	5,228	165,393
2026	195,409	6,490	5,228	183,691
2027	211,666	6,490	5,228	199,948
Total	1,055,335	45,429	36,596	973,310
Average over the Crediting Period	150,762	6,490	5,228	139,044

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	GWP _{CH4}
Data unit	t CO ₂ e/t CH ₄
Description	Global warming potential of CH ₄
Source of data	IPCC Fifth Assessment Report: Climate Change 2013 ⁸⁶
Value applied:	28
Justification of choice of data or description of measurement methods and procedures applied	Default value IPCC Fifth Assessment Report (AR5)
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	RATE _{COMPLIANCE,t}
Data unit	Fraction
Description	Rate of compliance with a regulatory requirement to implement the alternative waste treatment t implemented in the project activity
Source of data	Official Report – Solid Waste Management Plan published by MoEU
Value applied:	0
Justification of choice of data or description of measurement	There is no regulation in Turkey to mandate the implementation of alternative waste treatment options for anaerobic digestion and mechanical treatment employed for the project activity.

methods and procedures applied	
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	GWP _{N2O}
Data unit	t CO ₂ e/t N ₂ O
Description	Global Warming Potential of N ₂ O
Source of data	IPCC Fifth Assessment Report: Climate Change 2013 (AR5)
Value applied:	265
Justification of choice of data or description of measurement methods and procedures applied	Default value IPCC Fifth Assessment Report (AR5)
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	EF _{CO2,BL,CG}
Data unit	t CO ₂ /TJ
Description	Emission factor of baseline fossil fuel used in the cogeneration plant, as identified in the baseline scenario identification
Source of data	IPCC default value
Value applied:	58.3

Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	<i>Calculation of baseline emissions</i>
Comments	-

Data / Parameter	η_{cogen}
Data unit	ratio
Description	Efficiency of the cogeneration plant that would have been used in the absence of the project activity
Source of data	Manufacturers' efficiency value
Value applied:	84.2%
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	Φ_{default}
Data unit	-
Description	Default value for the model correction factor to account for model uncertainties
Source of data	Methodological Tool 04 Emissions from solid waste disposal sites Version 08.0

Value applied:	0.85						
Justification of choice of data or description of measurement methods and procedures applied	For baseline emissions: refer to the table below to identify the appropriate factor based on the application of the tool (A or B) and the climate where the SWDS is located. The project activity applies Application B, and the project area is located in relatively humid conditions. Default values for the model correction factor <table><tr><td></td><td>Humid/wet conditions</td><td>Dry conditions</td></tr><tr><td>Application B</td><td>0.85</td><td>0.80</td></tr></table>		Humid/wet conditions	Dry conditions	Application B	0.85	0.80
	Humid/wet conditions	Dry conditions					
Application B	0.85	0.80					
Purpose of Data	Calculation of baseline emissions						
Comments	The table above is applicable to Option 1 in the procedure “Determining the model correction factor (ϕ_y)”						

Data / Parameter	OX
Data unit	-
Description	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
Source of data	Based on an extensive review of published literature on this subject, including the IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied:	0.1
Justification of choice of data or description of measurement methods and procedures applied	Default value given in Methodological Tool 04 Emissions from solid waste disposal sites Version 08.0
Purpose of Data	Calculation of baseline emissions
Comments	When methane passes through the top layer, part of it is oxidized by methanotrophic bacteria to produce CO ₂ . The oxidation factor represents the proportion of methane that is oxidized to CO ₂ . This should be distinguished from the methane correction factor (MCF), which is to account for the situation that ambient air might intrude

	into the SWDS and prevent methane from being formed in the upper layer of SWDS
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Data / Parameter	F
Data unit	-
Description	Fraction of methane in the SWDS gas (volume fraction)
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied:	0.5
Justification of choice of data or description of measurement methods and procedures applied	Default value given in Methodological Tool 04 Emissions from solid waste disposal sites Version 08.0
Purpose of Data	Calculation of baseline emissions
Comments	Upon biodegradation, organic material is converted to a mixture of methane and carbon dioxide

Data / Parameter	DOC _{f,default}
Data unit	Weight fraction
Description	Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposes in the SWDS
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied:	0.5
Justification of choice of data or description of measurement methods and procedures applied	Default value given in Methodological Tool 04 Emissions from solid waste disposal sites Version 08.0

Purpose of Data	Calculation of baseline emissions
Comments	This factor reflects that some degradable organic carbon does not degrade, or degrades very slowly, in the SWDS. This default value can only be used for: (a) Application A; or (b) Application B if the tool is applied to MSW.

Data / Parameter	MCF _{default}
Data unit	-
Description	Methane correction factor
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied:	
Justification of choice of data or description of measurement methods and procedures applied	Default value given in Methodological Tool 04 Emissions from solid waste disposal sites Version 08.0 As per Tool 04, in case the SWDS does not have a water table above the bottom of the SWDS and in case of application A, then select the applicable value from the following: (a) 1.0 for anaerobic managed solid waste disposal sites. These must have controlled placement of waste (i.e. waste directed to specific deposition areas, a degree of control of scavenging and a degree of control of fires) and will include at least one of the following: (i) cover material; (ii) mechanical compacting; or (iii) levelling of the waste;
Purpose of Data	Calculation of baseline emissions
Comments	MCF accounts for the fact that unmanaged SWDS produce less methane from a given amount of waste than managed SWDS, because a larger fraction of waste decomposes aerobically in the top layers of unmanaged SWDS

Data / Parameter	DOC _j
Data unit	-
Description	Fraction of degradable organic carbon in the waste type j (weight fraction)

Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Tables 2.4 and 2.5)														
Value applied:	Default values for DOC_j <table> <tr> <th>Waste type j</th><th>DOC_j (% wet waste)</th></tr> <tr> <td>Wood and wood products</td><td>43</td></tr> <tr> <td>Pulp, paper and cardboard (other than sludge)</td><td>40</td></tr> <tr> <td>Food, food waste, beverages and tobacco (other than sludge)</td><td>15</td></tr> <tr> <td>Textiles</td><td>24</td></tr> <tr> <td>Garden, yard and park waste</td><td>20</td></tr> <tr> <td>Glass, plastic, metal, other inert waste</td><td>0</td></tr> </table>	Waste type j	DOC_j (% wet waste)	Wood and wood products	43	Pulp, paper and cardboard (other than sludge)	40	Food, food waste, beverages and tobacco (other than sludge)	15	Textiles	24	Garden, yard and park waste	20	Glass, plastic, metal, other inert waste	0
Waste type j	DOC_j (% wet waste)														
Wood and wood products	43														
Pulp, paper and cardboard (other than sludge)	40														
Food, food waste, beverages and tobacco (other than sludge)	15														
Textiles	24														
Garden, yard and park waste	20														
Glass, plastic, metal, other inert waste	0														
Justification of choice of data or description of measurement methods and procedures applied	Default value														
Purpose of Data	Calculation of baseline emissions														
Comments	The percentages listed in table above are based on wet waste basis which are concentrations in the waste as it is delivered to the SWDS. The IPCC Guidelines also specify DOC values on a dry waste basis, which are the concentrations after complete removal of all moist from the waste, which is not believed practical for this situation														

Data / Parameter	k_j
Data unit	1/yr
Description	Decay rate for the waste type j
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Table 3.3)

Value applied:	Default values for the decay rate (k_i)					
	Waste type /		Boreal and Temperate (MAT$\leq$20°C)		Tropical (MAT>20°C)	
			Dry (MAP/PET <1)	Wet (MAP/PET >1)	Dry (MAP < 1000mm)	Wet (MAP > 1000mm)
	Slowly degrading	Pulp, paper, cardboard (other than sludge), textiles	0.04	0.06	0.045	0.07
		Wood, wood products and straw	0.02	0.03	0.025	0.035
Moderately degrading	Other (non-food) organic putrescible garden and park waste		0.05	0.10	0.065	0.17
	Food, food waste, sewage sludge, beverages and tobacco		0.06	0.185	0.085	0.40
Justification of choice of data or description of measurement methods and procedures applied	Default value given in Methodological Tool 04 Emissions from solid waste disposal sites Version 08.0 As per Tool 04, MAT – mean annual temperature, MAP – Mean annual precipitation, PET – potential evapotranspiration. MAP/PET is the ratio between the mean annual precipitation and the potential evapotranspiration.					
Purpose of Data	Calculation of baseline emissions					
Comments	Document in the CDM-PDD the climatic conditions at the SWDS site (temperature, precipitation and, where applicable, evapotranspiration). Use long-term averages based on statistical data, where available					

Data / Parameter	R_u
Data unit	Pa.m ³ /kmol.K
Description	Universal ideal gases constant
Source of data	Tool 08: Tool to determine the mass flow of a greenhouse gas in a gaseous stream Version 03.0
Value applied:	8,314
Justification of choice of data or description of measurement methods and procedures applied	Default value given in Methodological Tool 08 Tool to determine the mass flow of a greenhouse gas in a gaseous stream Version 03.0

Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	MM _i														
Data unit	kg/kmol														
Description	Molecular mass of greenhouse gas i														
Source of data	Tool 08: Tool to determine the mass flow of a greenhouse gas in a gaseous stream Version 03.0														
Value applied:	<table><tr><th>Compound</th><th>Structure</th><th>Molecular mass (kg / kmol)</th></tr><tr><td>Carbon dioxide</td><td>CO₂</td><td>44.01</td></tr><tr><td>Methane</td><td>CH₄</td><td>16.04</td></tr><tr><td>Nitrous oxide</td><td>N₂O</td><td>44.02</td></tr></table>			Compound	Structure	Molecular mass (kg / kmol)	Carbon dioxide	CO ₂	44.01	Methane	CH ₄	16.04	Nitrous oxide	N ₂ O	44.02
Compound	Structure	Molecular mass (kg / kmol)													
Carbon dioxide	CO ₂	44.01													
Methane	CH ₄	16.04													
Nitrous oxide	N ₂ O	44.02													
Justification of choice of data or description of measurement methods and procedures applied	Default values given in Methodological Tool 08 Tool to determine the mass flow of a greenhouse gas in a gaseous stream Version 03.0														
Purpose of Data	Calculation of baseline emissions														
Comments	-														

Data / Parameter	P _n
Data unit	Pa
Description	Total pressure at normal conditions
Source of data	Tool 08: Tool to determine the mass flow of a greenhouse gas in a gaseous stream Version 03.0
Value applied:	101,325 Pa

Justification of choice of data or description of measurement methods and procedures applied	Default values given in Methodological Tool 08 Tool to determine the mass flow of a greenhouse gas in a gaseous stream Version 03.0
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	T_n
Data unit	K
Description	Temperature at normal conditions
Source of data	Tool 08: Tool to determine the mass flow of a greenhouse gas in a gaseous stream Version 03.0
Value applied:	273.15 K
Justification of choice of data or description of measurement methods and procedures applied	Default values given in Methodological Tool 08 Tool to determine the mass flow of a greenhouse gas in a gaseous stream Version 03.0
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	ρ_{CH_4}
Data unit	t CH ₄ / m ³ CH ₄
Description	Density of methane at normal conditions

Source of data	'Thermophysical properties of fluids. II. Methane, Ethane, Propane, Isobutane and Normal Butane' by B.A. Younglove, J.F. Ely Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0
Value applied:	0.00067
Justification of choice of data or description of measurement methods and procedures applied	Default values given in Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0
Purpose of Data	Calculation of project emissions
Comments	corrected to reference conditions

Data / Parameter	$EF_{CH_4, default}$
Data unit	t CH ₄ leaked / t CH ₄ produced
Description	Default emission factor for the fraction of CH ₄ produced that leaks from the anaerobic digester
Source of data	IPCC (2006), Flesch et al. (2011) and Kurup (2003)
Value applied:	0.028
Justification of choice of data or description of measurement methods and procedures applied	As per Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0 Digesters with steel or lined concrete or fiberglass digesters and a gas holding system (egg shaped digesters) and monolithic construction shall use the default of 0.028
Purpose of Data	Calculation of project emissions
Comments	Applicable to the step "Determination of project emissions of methane from the anaerobic digester"

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Combined margin emission factor for the grid in year y
Source of data	Republic of Turkey Ministry of Energy in Emission Factor 2020
Value applied:	0.5552
Justification of choice of data or description of measurement methods and procedures applied	Calculated ex ante and fixed for the 7-year renewable crediting period.
Purpose of Data	Determination of baseline emissions
Comments	-

Data / Parameter	$TDL_{j,y}$
Data unit	-
Description	Average technical transmission and distribution losses for providing electricity to source j or k in year y
Source of data	Methodological tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0)
Value applied:	3%
Justification of choice of data or description of measurement methods and procedures applied	According to the tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0), the project belongs to the case of Scenario A, use as default values of 3% for project emission since electricity consumption by all project electricity consumption sources is smaller than the electricity

	consumption of all baseline electricity consumption sources, and 3% for baseline emissions.
Purpose of Data	Calculation of project emissions from electricity consumption
Comments	-

Data / Parameter	$EF_{N_2O, D, j}$
Data unit	kg N ₂ O-N/kg N
Description	Direct N ₂ O emission factor for the treatment system j of the manure management system
Source of data	Obtained from 2019 IPCC Refinement, Table 10.21, p. 10.91 (uncovered anaerobic lagoon)
Value applied:	0.00
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Used in baseline direct N ₂ O emissions calculations.
Comments	-

Data / Parameter	$EF_{N_2O, ID}$
Data unit	kg N ₂ O-N/kg NH ₃ -N and NO _x -N
Description	Indirect N ₂ O emission factor for N ₂ O emissions from atmospheric deposition of nitrogen on soils and water surfaces
Source of data	Obtained from 2019 IPCC Refinement, Table 11.3, p. 11.26

Value applied:	0.010
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Used in baseline and project indirect N ₂ O emissions calculations
Comments	-

Data / Parameter	F_{gas}
Data unit	Fraction
Description	Fraction of N lost due to volatilization
Source of data	Obtained from 2019 IPCC Refinement, Table 11.3, p. 11.26
Value applied:	0.21
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Used in baseline and project N ₂ O emissions calculations
Comments	-

Data / Parameter	MCF_j
Data unit	-
Description	Methane conversion factor for the baseline AWMS _j

Source of data	Obtained from 2019 IPCC Refinement, Table 10.17, p. 67 (uncovered anaerobic lagoon for 17.9°C, warm temperate moist)
Value applied:	0.73
Justification of choice of data or description of measurement methods and procedures applied	Default value. Uncovered anaerobic lagoon, Value for 17.9°C.
Purpose of Data	Used in project emission/baseline calculations
Comments	A conservativeness factor should be applied by multiplying MCF values (estimated as per the above bullet) with a value of 0.94 to account for the 20 per cent uncertainty in the MCF values as reported by IPCC

Data / Parameter	VS _{default}
Data unit	kg -dm/animal/day
Description	The default value for the volatile solid excretion per day on a dry-matter basis for a defined livestock population
Source of data	Obtained from 2019 IPCC Refinement, Table 10.13A p. 10.55 Middle East
Value applied:	10.7 (1000 kg animal mass/day) for dairy cattle 14.1 (1000 kg animal mass/day) for other cattle
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Used in project emission/baseline calculations
Comments	-

Data / Parameter	NEX _{IPCCdefault}
Data unit	kg N/animal/year
Description	The default value for the nitrogen excretion per head of a defined livestock population
Source of data	Obtained from 2019 IPCC Refinement, Table 10.19, p. 10.83, Middle East
Value applied:	For dairy cattle: 0.50 kg N (1000 kg animal mass)/day For other cattle: 0.55 kg N (1000 kg animal mass)/day
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Used in baseline direct N ₂ O emissions calculations.
Comments	This parameter is used to estimate NEX _{LT,y} in appendix 2 of ACM0010

5.2 Data and Parameters Monitored

Data / Parameter	EC _{t,y}
Data unit	MWh
Description	Electricity consumption of electricity generated in an on-site fossil fuel fired power plant or from the grid as a result of the alternative waste treatment process t in year y
Source of data	Electricity meters
Description of measurement	Sources of consumption include the operation of the alternative waste treatment process, on-site processing or

methods and procedures applied	management of the feedstock or products associated with the treatment process. Electricity consumption is monitored for all activities included in the project boundary, associated with the treatment process.
Frequency of monitoring/recording	Continuous
Value applied:	3,792 (Obtained from the Capacity Report of the Project Activity)
Monitoring equipment	Electricity meters
QA/QC procedures applied	Electricity meters used in the Project Activity are subject to regular (in accordance with stipulation of the meter supplier) maintenance and testing to ensure accuracy. The readings are also double checked against invoices provided by TEIAS
Purpose of data	This parameter is required for calculating project emissions from electricity consumption due to waste treatment under the project activity
Calculation method	No calculation is needed measurement values are used.
Comments	EC _{t,y} excludes consumption of any electricity generated by the project activity

Data / Parameter	EG _{t,y}
Data unit	MWh
Description	Electricity generated by the alternative waste treatment process t and exported to the grid or displacing fossil fuel fired power only and/or cogeneration captive energy generation in year y
Source of data	Electricity meters
Description of measurement methods and procedures applied	

Frequency of monitoring/recording	Continuous
Value applied:	37,307.6 (Obtained from the Capacity Report of the Project Activity)
Monitoring equipment	Electricity meters
QA/QC procedures applied	Electricity meters are subject to regular (in accordance with stipulation of the meter supplier) maintenance and testing to ensure accuracy.
Purpose of data	This parameter is required for calculating baseline emissions from electricity generation due to waste treatment under the project activity.
Calculation method	No calculation is needed measurement values are used.
Comments	-

Data / Parameter	$HG_{PJ,y}$
Data unit	TJ
Description	Quantity of heat supplied by the project activity displacing baseline heat generation by a fossil fuel boiler or air heater in year y (TJ)
Source of data	Steam meters
Description of measurement methods and procedures applied	In case of steam meter: the enthalpy of steam and feed water will be determined at measured temperature and pressure and the enthalpy difference will be multiplied with quantity measured by steam meter.
Frequency of monitoring/recording	Monthly, aggregated yearly
Value applied:	113.8730 (=13,660,714*5.5*0.421*(3.6*10 ⁻⁶))
Monitoring equipment	Steam meters

QA/QC procedures applied	In case of monitoring of steam, it will be calibrated for pressure and temperature of steam at regular intervals. The meter shall be subject to regular maintenance and testing to ensure accuracy
Purpose of data	To calculate baseline emissions resulting from heat generation
Calculation method	No calculation is needed measurement values are used.
Comments	The dedicated quantity of thermal energy generated for heat supply or cogeneration by the project activity if included

Data / Parameter	Q _{RDF_SB,COM,y}
Data unit	T
Description	Quantity of RDF/SB exported off-site with potential to be combusted in year y
Source of data	Project site
Description of measurement methods and procedures applied	Sale invoices of the RDF/SB should be kept at the project site. They shall contain customer contact details, physical location of delivery, type, amount (in tons) and purpose of RDF/SB (use as fuel or as material in furniture, etc.). A list of customers and delivered SD amount shall be kept at the project site
Frequency of monitoring/recording	Weekly
Value applied:	2,400
Monitoring equipment	Sale invoices of the RDF
QA/QC procedures applied	-
Purpose of data	to calculate leakage emissions associated with RDF production

Calculation method	Measured
Comments	-

Data / Parameter	$Q_{RDF_SB,y}$
Data unit	T
Description	Quantity of RDF/SB produced in year y
Source of data	Project participants
Description of measurement methods and procedures applied	Weighbridge
Frequency of monitoring/recording	Annually
Value applied:	2,400
Monitoring equipment	Weighbridge
QA/QC procedures applied	Weighbridge will be subject to periodic calibration (in accordance with stipulation of the weighbridge supplier)
Purpose of data	to calculate leakage emissions associated with RDF production
Calculation method	Measured
Comments	-

Data / Parameter	$EF_{CO_2,RDF_SB,y}$
Data unit	t CO ₂ /GJ
Description	Weighted average CO ₂ emission factor for RDF/SB in year y

Source of data	IPCC default values at the upper/lower limit of the uncertainty at a 95 per cent confidence interval as provided in table 1.4 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Description of measurement methods and procedures applied	The default value for Municipal Wastes (non-biomass fraction)
Frequency of monitoring/recording	Any future revision of the IPCC Guidelines will be taken into account
Value applied:	0.121
Monitoring equipment	Default Value
QA/QC procedures applied	Default Value
Purpose of data	This parameter is required for the procedure to calculate leakage emissions for the combustion of RDF/SB outside the project boundary
Calculation method	Default Value
Comments	The upper limit is chosen to be conservative since project emissions are calculated.

Data / Parameter	$NCV_{RDF_SB,y}$
Data unit	GJ/ton
Description	The weighted average net calorific value of RDF/SB in year y
Source of data	IPCC default values at the upper/lower limit of the uncertainty at a 95 per cent confidence interval as provided in table 1.2 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Description of measurement	The default value for Municipal Wastes (non-biomass fraction)

methods and procedures applied	
Frequency of monitoring/recording	Any future revision of the IPCC Guidelines will be taken into account
Value applied:	18
Monitoring equipment	Default Value
QA/QC procedures applied	Default Value
Purpose of data	This parameter is required for the procedure to calculate leakage emissions for the combustion of RDF/SB outside the project boundary
Calculation method	Default Value
Comments	The upper limit is chosen to be conservative since project emissions are calculated.

Data / Parameter	f_y
Data unit	-
Description	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
Source of data	the maximum value from the following: (a) contract or regulation requirements specifying the amount of methane that must be destroyed/used (if available) and (b) historic data on the amount captured
Description of measurement methods and procedures applied	Annually checked according to national law and regulations.
Frequency of monitoring/recording	For application B: Annually

Value applied:	0
Monitoring equipment	
QA/QC procedures applied	-
Purpose of data	Determination of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	W _x or W _i																	
Data unit	t																	
Description	Total amount of waste disposed in a SWDS in year x or month i																	
Source of data	Measurements by project participants																	
Description of measurement methods and procedures applied	Measure on wet basis																	
Frequency of monitoring/recording	Continuously, aggregated at least annually for year x																	
Value applied:	<table><tr><th>Year</th><th>Total (tonnes)</th></tr><tr><td>2021</td><td>268,434</td></tr><tr><td>2022</td><td>269,936</td></tr><tr><td>2023</td><td>271,400</td></tr><tr><td>2024</td><td>273,015</td></tr><tr><td>2025</td><td>274,399</td></tr><tr><td>2026</td><td>276,027</td></tr><tr><td>2027</td><td>277,422</td></tr></table>		Year	Total (tonnes)	2021	268,434	2022	269,936	2023	271,400	2024	273,015	2025	274,399	2026	276,027	2027	277,422
Year	Total (tonnes)																	
2021	268,434																	
2022	269,936																	
2023	271,400																	
2024	273,015																	
2025	274,399																	
2026	276,027																	
2027	277,422																	
Monitoring equipment	Weighbridge on the site																	
QA/QC procedures applied	Weighbridge is subject to periodic calibration (in accordance with of the manufacturer's specifications)																	

Purpose of data	Calculation of baseline emissions
Calculation method	Measured
Comments	For application B

Data / Parameter	$V_{t,wb}$
Data unit	m ³ wet gas/h
Description	Volumetric flow of the gaseous stream in time interval t on a wet basis
Source of data	Flow meters
Description of measurement methods and procedures applied	Volumetric flow measurement refers to the actual pressure and temperature. Instruments with recordable electronic signal (analogical or digital) are conducted
Frequency of monitoring/recording	Continuous
Value applied:	13,660,714 m ³ /yr (Obtained from the capacity report)
Monitoring equipment	Flow meters
QA/QC procedures applied	Periodic calibration against a primary device provided by an independent accredited laboratory is mandatory for the project activity. Calibration and frequency of calibration is done according to manufacturer's specifications
Purpose of data	<i>To calculate</i> project emissions
Calculation method	Measured
Comments	This parameter will be monitored in Options B and C

Data / Parameter	$V_{i,t,wb}$
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Data unit	m ³ gas i/m ³ wet gas
Description	Volumetric fraction of greenhouse gas i in a time interval t on a wet basis
Source of data	Flow meters
Description of measurement methods and procedures applied	Calculated based on the dry basis analysis plus water concentration measurement or continuous in-situ analyzers if not specified in the underlying methodology
Frequency of monitoring/recording	Continuous
Value applied:	50% (conservative value as per ACM0010)
Monitoring equipment	Flow meters
QA/QC procedures applied	Calibration should include zero verification with an inert gas (e.g. N ₂) and at least one reading verification with a standard gas (single calibration gas or mixture calibration gas). All calibration gases must have a certificate provided by the manufacturer and must be under their validity period.
Purpose of data	<i>To calculate project emissions</i>
Calculation method	Measured
Comments	This parameter will be monitored in Options C and F

Data / Parameter	T _t
Data unit	K
Description	Temperature of the gaseous stream in time interval t
Source of data	Thermocouple
Description of measurement methods and procedures applied	Instrument with recordable electronic signal (analogical or digital) is used.

Frequency of monitoring/recording	Continuous
Value applied:	will be monitored ex post
Monitoring equipment	Thermocouple
QA/QC procedures applied	Periodic calibration against a primary device provided by an independent accredited laboratory is mandatory. Calibration and frequency of calibration are according to the manufacturer's specifications
Purpose of data	To calculate project emissions
Calculation method	Measured
Comments	Provided all parameters are converted to normal conditions during the monitoring process, this parameter may not be needed except for moisture content determination. Therefore it should be metered only when performing such measurements (with the same frequency). However, if the applicability condition related to the gaseous stream flow temperature being below 60°C is adopted, this parameter must be monitored continuously to assure the applicability condition is met

Data / Parameter	Pt
Data unit	Pa
Description	Pressure of the gaseous stream in time interval t
Source of data	pressure transducer
Description of measurement methods and procedures applied	Instruments with recordable electronic signal (analogical or digital) is used
Frequency of monitoring/recording	Continuous

Value applied:	will be monitored ex post
Monitoring equipment	pressure transducer
QA/QC procedures applied	Periodic calibration against a primary device must be performed periodically and records of calibration procedures must be kept available as well as the primary device and its calibration certificate. Pressure transducers (either capacitive or resistive) must be calibrated monthly
Purpose of data	To calculate project emissions
Calculation method	Measured
Comments	Provided all parameters are converted to normal conditions during the monitoring process, this parameter may not be needed except for moisture content determination and therefore it should be metered only when performing such measurements (with same frequency)

Data / Parameter	$F_{CH_4,m}$
Data unit	kg
Description	Mass flow of methane in the residual gaseous stream in the minute m
Source of data	Calculated based on the flow of biogas and the concentration of methane in the biogas that will be sent to the flare
Description of measurement methods and procedures applied	Measured by a flow meter and a gas analyser. Data to be aggregated monthly and yearly. Measured by a flow meter and a gas analyser. Data to be aggregated monthly and yearly
Frequency of monitoring/recording	Continuous
Value applied:	26,800
Monitoring equipment	Measured by a flow meter and a gas analyser.

QA/QC procedures applied	Flow meters are subject to regular (in accordance to the manufacturer) maintenance and testing to ensure accuracy.
Purpose of data	To calculate project emissions
Calculation method	Calculated based on the flow of biogas and the concentration of methane in the biogas that will be sent to the flare
Comments	-

Data / Parameter	B _{0,LT}							
Data unit	m ³ CH ₄ /kg dm							
Description	Maximum methane producing potential of the volatile solid generated by animal type LT							
Source of data	2019 IPCC Refinement, Table 10.16, p. 10.66, Eastern Europe							
Description of measurement methods and procedures applied	Default Value							
Frequency of monitoring/recording	Annually							
Value applied:	<table><tr><td>Dairy Cattle</td><td>0.24</td></tr><tr><td>Other Cattle</td><td>0.17</td></tr><tr><td>Poultry</td><td>0.36</td></tr></table>		Dairy Cattle	0.24	Other Cattle	0.17	Poultry	0.36
Dairy Cattle	0.24							
Other Cattle	0.17							
Poultry	0.36							
Monitoring equipment	Default Value							
QA/QC procedures applied	-							
Purpose of data	To calculate Baseline Methane Emissions due to animal manure usage							
Calculation method	Default value							
Comments	-							

Data / Parameter	Type			
Data unit	-			
Description	Type of barn and AWMS			
Source of data	Project proponents			
Description of measurement methods and procedures applied	Records kept by the Project Proponent will be used to determine the types			
Frequency of monitoring/recording	Annually			
Value applied:	<table><tr><td>Dairy Cattle</td></tr><tr><td>Other Cattle</td></tr><tr><td>Poultry</td></tr></table>	Dairy Cattle	Other Cattle	Poultry
Dairy Cattle				
Other Cattle				
Poultry				
Monitoring equipment	-			
QA/QC procedures applied	Agreements signed with the farmers will be used for cross check.			
Purpose of data	To confirm that the implementation of project is being carried out as per design			
Calculation method	-			
Comments	-			

Data / Parameter	T
Data unit	°C
Description	Annual average ambient temperature at project site
Source of data	Project proponents
Description of measurement	Plant's log

methods and procedures applied	
Frequency of monitoring/recording	Monthly
Value applied:	17.9
Monitoring equipment	-
QA/QC procedures applied	Cross-Check with the Annual average ambient temperature at project city from public website monthly
Purpose of data	Calculation of baseline emissions
Calculation method	Measured
Comments	Used to select the annual MCF from IPCC

Data / Parameter	$N_{da,LT}$
Data unit	Number
Description	Number of days animal of type LT is alive in the farm in the year y
Source of data	Project proponents
Description of measurement methods and procedures applied	Plant's log
Frequency of monitoring/recording	Monthly
Value applied:	365
Monitoring equipment	-
QA/QC procedures applied	Agreements signed with the farmers will be used for cross check

Purpose of data	Baseline/emission reductions calculations
Calculation method	-
Comments	-

Data / Parameter	$N_{p,LT}$	
Data unit	Number	
Description	Number of animals of type LT produced annually for the year y	
Source of data	Project proponents	
Description of measurement methods and procedures applied	Plant's log	
Frequency of monitoring/recording	Monthly	
Value applied:	Other Cattle	600
	Dairy Cattle	2,345
	Poultry	3,500
Monitoring equipment	-	
QA/QC procedures applied	Agreements signed with the farmers will be used for cross check	
Purpose of data	Baseline/emission reductions calculations	
Calculation method	-	
Comments	-	

Data / Parameter	$MS\%_j$
Data unit	Fraction

Description	Fraction of manure handled in system j in the project activity
Source of data	Project proponents
Description of measurement methods and procedures applied	Plant's log
Frequency of monitoring/recording	Annually
Value applied:	100%
Monitoring equipment	-
QA/QC procedures applied	100% is the maximum value and conservative.
Purpose of data	Baseline/emission reductions calculations
Calculation method	-
Comments	-

Data / Parameter	$NEX_{LT,y}$
Data unit	kg N/animal/year
Description	Annual average nitrogen excretion per head of a defined livestock population estimated as described in appendix 2 of ACM0010
Source of data	2019 IPCC Refinement, Table 10.19, Middle East, Mean
Description of measurement methods and procedures applied	Default value
Frequency of monitoring/recording	Annually

Value applied:	Other Cattle	0.55
	Dairy Cattle	0.50
	Poultry	1.29
Monitoring equipment	Default value	
QA/QC procedures applied	Default value	
Purpose of data	to calculate baseline N ₂ O emissions due to manure usage	
Calculation method	Default value	
Comments	When using equation 2 in appendix 2, please refer to above guidance for estimating W_{site}	

Data / Parameter	nd_y
Data unit	Number
Description	Number of days treatment plant was operational in year y
Source of data	Project proponents
Description of measurement methods and procedures applied	-
Frequency of monitoring/recording	Daily
Value applied:	300 (Obtained from the Capacity Report)
Monitoring equipment	SCADA
QA/QC procedures applied	The SCADA system employed at the plant will be used for crosscheck to see how many days the treatment plant is operational.
Purpose of data	Baseline/emission reductions calculations
Calculation method	-

Comments

-

5.3 Monitoring Plan

The monitoring methodology is based on directly measuring the biogas produced in the digesters, the biogas sent to the flare unit, and the electricity generated in the engines, as shown in the table below. The monitoring plan continuously measures the quantity and quality of biogas generated and flared. The main variables that need to be determined are the quantity of Volumetric flow of the gaseous stream in the hour h on a dry basis ($V_{t,db}$) and the quantity of methane used to generate electricity. The methodology also measures the energy generated using biogas ($EC_{BL,y}$).

From the monitoring methodology, it could be seen that there are the following main variables to be measured:

Table 4: Summary of Monitoring Plan

Parameter	Description
$V_{t,db}$	The volumetric flow of the gaseous stream in the hour h on a dry basis (Nm ³)
$F_{CH_4,sentflare,y}$	Amount of methane in LFG which is sent to the flare in year y
$F_{CH_4,EL,y}$	Amount of methane in LFG which is sent to the genset for electricity generation in year y
$V_{CH_4,t,db}$	The volumetric flow of CH ₄ in time interval on a dry basis (Nm ³)
T	The temperature of the biogas
P	The pressure of the gaseous stream in the hour h
W_{CH_4}	Methane fraction in the biogas
$EC_{BL,y}$	Net electricity delivered to the grid

Responsibilities for data processing and management lie with Biosun. Therefore, they established a VER team. This team is responsible for monitoring all data required to estimate emission reductions. Life Enerji also assists VER Team concerning the monitoring aspects of the Project. The plant manager has the main responsibility for collecting and archiving the data. According to the monitoring plan, qualified technicians monitor and record the data. All the technicians receive proper training to ensure that they understand their specific tasks and equipment handling. The records are double-checked by the General Manager of the Proposed Project, who is responsible for the accuracy and frequency of the measurements.

EPIAS stores the power generation data. The website of EPIAS (<https://www.epias.com.tr/uzlastirma/aylik-uzlastirma-bildirimsureci/>) is accessible by the Project owner with their unique user ID and password. Once accessed, the Project owners can retrieve electricity generation and consumption data. The Project owner uses the same data for invoicing purposes. The electricity generation data are reported on a monthly basis.

Data Processing and collection

The monitoring equipment and other plant equipment are connected to a Programmable Logic Controller (PLC), which collects all the monitoring data and sends them to the SCADA system so that the operator can read the main parameters in real-time. The PLC archives reading values in a database which is accessible through the SCADA interface. Monitored data can be read and downloaded with the SCADA system for reporting purposes. Moreover, these electronic records are backed-up in a separate computer. The main source for landfill gas data is the Project activity's PLC-SCADA system. The system automatically reads measurement devices and records data on hard drives. The PLC system records meter readings at pre-determined intervals, as specified in the CDM monitoring manual. This data is used to calculate the total emission reductions.

1) Flow measurements

Gas flow is measured by flow meters, including the flow into the flare, generators and the total flow. These parameters are required to be normalised at 0°C and 1 atm, using actual gas temperature and pressure data to be measured with temperature and pressure transmitters. These in-built transmitters are integrated into each flow meter and equipped with a flow computer so that the values sent to the SCADA system have already been normalised (converted to normal cubic meters). Therefore, temperature and pressure data at each flowmeter do not need to be separately monitored. Methane content in the gas is measured by a gas analyser connected to the PLC-SCADA system.

Calibration: The flow sensors are calibrated according to the gas's specified temperature, pressure, and composition as per the manufacturer's recommendation. The equipment selected will allow dynamic compensation for these parameters, normalised to standard temperature, pressure, and gas composition. There will be a periodic verification according to the requirement of equipment specifications.

2) Gas Quality and efficiency of the flare

An enclosed flare unit determines methane destruction with an inbuilt flame detection unit. The project emissions from flaring and emission reductions from the destruction of biogas have been monitored at this point. The readings have been taken and recorded internally by compatible and proper measurement devices with high data quality. The PLC system automatically converts the measured parameters to dry- basis and normal conditions. The

instruments monitor parameters continuously. The emissions are calculated using the methodological tool "Project emissions from flaring" (v04.0).

The concentration of methane and oxygen in the gaseous stream and the exhaust gas of the flare are the parameters that are essential for the calculation of emission reductions and the safe and efficient operation of the system.

The concentration of methane and oxygen in the biogas stream is controlled by a common sample line installed in the main collection system piping and measured continuously by two separate analysers to measure methane, oxygen and carbon dioxide each. The sensors are designed to operate within specified temperature and pressure conditions.

The concentration of methane and oxygen in the exhaust gas stream is monitored by a common sample line installed in the upper section of the flare.

Calibration: Analysers are periodically calibrated according to the manufacturers and regulations on "Metering and Testing of Metering Systems"²¹ of the Ministry of Industry and Technology recommendation. Calibration equipment will provide +/- 1% accuracy by volume.

3) Electricity Generation:

The electricity generation data is stored by EPIAS, the financial settlement centre of TEİAŞ (the national grid operator). The website of EPIAS (<https://www.epias.com.tr/uzlastirma/>) is accessible by the Project owner with their ID used for control and checks and invoicing purposes. The electricity generation data is reported on a monthly basis. The procedure involves the following tasks:

- Accessing the website of EPIAS (<https://www.epias.com.tr/uzlastirma/>),
- Obtaining the electricity generation and consumption reports
- Store the data in electronic version on the computer of Biosun on the plant
- Issue invoices to TEİAŞ based on the data published in EPIAS.

A Backup Unit for Systematic Storage

i. Electricity Generation:

There are two measurement devices; a primary and a secondary (i.e. backup or check-meter) measurement device for quality assurance, to be used if the primary meter fails. Both the main and backup measurement devices measure and store the aggregate total electricity import and export for the whole Project in real-time. The electricity meters are under the responsibility of TEİAŞ, and they are located in the sub-station, which also belongs to TEİAŞ.

²¹ See, http://www.sanayi.gov.tr/download/osgm/olcu_aletleri_muayene_yonetmelik.zip (page 2)

The invoices are kept by the Project owner as hard copies. Furthermore, the EPIAS system stores the reports electronically, which is accessible to the Project owner whenever necessary. After the monthly reports are obtained from EPIAS, data are aggregated using a separate spreadsheet. The monthly generation and consumption data are entered into the spreadsheet, which calculates the emission reductions during the monitoring period.

ii. Methane Destruction:

Data Storage: The electronic monitoring system (PLC) will periodically archive the reading data. Written documents (e.g. equipment replacement protocols, accident logs, maintenance records, back-ups etc.) will be kept safely. Electronically backup of the data will be conducted on a daily basis. A hard copy backup of all relevant data will be printed out monthly. Calibration records for all instrumentation will be constantly collected and archived. All data and records required for verification are kept for two years after the project crediting period ends or the last issuance of VERs, whichever is later.

Calibration Procedures:

Maintenance and calibration of measurement devices: According to manufacturer specifications, all measurement devices will be purchased and maintained as specified in this report. All measurement devices that are used in monitoring will be subject to a quality control procedure that will include regular maintenance and calibration in agreement with legal and/or manufacturer requirements. According to the relevant regulation, "The Turkish Electricity Market Regulation Agency (EPDK) sets rules on the accuracy of electricity meters that are used by power plants feeding into the grid. The rules are part of the EPDK regulation 25056 from 22/03/200322. The table in Article 11 of the regulation specifies electricity meters of the accuracy class 0.5S for power plants between 10 MW and 100 MW and refers to compliance with the International Electrotechnical Commission's norm EN 60687. TEIAS, whose employees visit the plant for the meter readings monthly, is in charge of ensuring adherence to these rules. Calibration and maintenance procedures follow the requirements."

If any equipment malfunction or breakdown occurs, corrective actions are carried out in a timely manner to minimise the risk of emissions that are not intended. To ensure data quality, operational staff will be trained appropriately to take action effectively in such cases. The plant operator periodically inspects the plant to check visually if there are any apparent problems. In case of any findings, these are documented.

In case of an organisational change within the Project Plant, a qualified person will be assigned to carry out and manage the monitoring and verification procedures. In case of fire, earthquake or another similar emergency situation, the Project owner's data as backup stores will be used to maintain conservativeness with specific procedures. For extraordinary events where such backup data are unavailable, the amount of biogas consumed will be

²² http://www.epdk.gov.tr/documents/elektrik/mevzuat/teblig/elektrik/sayaclar_hakkinda/Elk_Tblg_Sayaclar.doc, page 3.

calculated using the power generation figures and engines' full-load electrical efficiencies by using conservative values and other assumptions to maintain conservativeness. Regarding the project emissions from flaring, data recording and aggregation and procedures for extraordinary events are the same as described above (i.e. the methane destruction component).

Monitoring will be carried out following the procedures set by the applicable methodology and all applicable tools of the above-explained parameters. The monitoring consists of:

Responsibilities for monitoring

The project owner is responsible for the installed equipment operation and maintenance. The project owner is also responsible for the administration of the data, setting up a VER team who will be responsible for monitoring all data required to estimate emission reductions.

Registration of the monitored data

The Ödemis ISWMP Project VER-Team will be responsible for monitoring equipment for quality assurance. The data measured by the monitoring equipment will be stored and will be processed into a monitoring report, which the project owner will submit. All the monitored data will be stored during the crediting period and for at least two years after the end of the crediting period, whichever occurs later.

Corrective actions and emergency preparedness

The VER-Team will regularly check the monitoring system for errors. In the case of errors, corrective actions will be undertaken by the VER-Team, or if required, by the monitoring equipment supplier.

QA/QC procedure

Strong quality assurance and quality control procedures will be taken to monitor the equipment and data collection. Equipment and facilities will be subject to a regular maintenance and testing regime to ensure accuracy following the supplier's manual. In case of data deficiency or correct data reading, the data will be cross-checked with other parameters and data leading to the most conservative emission reductions will be considered for calculating the emission reductions.

Measuring of exported and imported electricity

The main meter for monitoring the electricity import and export is at the grid connection. Monthly protocols based on the readings confirmed by TEİAŞ and the project participant are prepared by the end of each month and stored. These protocols are also the basis of invoicing for the electricity sales from the proposed project activity. The electricity meters are

bi-directional and will measure both the imported and exported electricity from and to the grid.

6 ACHIEVED GHG EMISSION REDUCTIONS AND REMOVALS

6.1 Data and Parameters Monitored

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

6.2 Baseline Emissions

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

6.3 Project Emissions

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

6.4 Leakage

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

6.5 Net GHG Emission Reductions and Removals

After the Project completes the Listing in the Pipeline, this section will be provided under the validation and verification phase.

APPENDIX X: <TITLE OF APPENDIX>

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.